

Belief Geometry on the Random Hierarchy Model

A small transformer partially encodes — and causally uses — the exact tree posterior, probed against ground truth

simplex-rhm-belief · run of 29 June 2026 · all numbers from the recorded pipeline output

Abstract. We reproduce, in miniature, the Simplex belief-geometry result on a hierarchical generative process. We train a 2-layer GPT-2 decoder ($\approx 399\text{k}$ parameters) by next-token prediction on samples from a Random Hierarchy Model (RHM; arity $s = 2$, depth $L = 3$, vocabulary $v = 8$), a grammar small enough (1024 equiprobable trees) to enumerate exactly. We compute the **exact** Bayesian posterior over the tree’s hidden latents by sum-product belief propagation, verified against brute-force enumeration to $< 10^{-6}$. The trained model reaches a test cross-entropy of 0.88 nats, closing $\approx 89\%$ of the gap between the uniform baseline (2.08) and the Bayes-optimal floor (0.73) — it has effectively learned the posterior predictor. A single global linear probe partially recovers this posterior from the residual stream: held-out $R^2 = 0.38$ for the root class versus ≈ 0 for a shuffled control — a real but noisy affine image, not an exact reconstruction — with decodability accumulating monotonically across depth ($-0.00 \rightarrow 0.15 \rightarrow 0.38$) and the belief readout **blooming** outward from the prior toward simplex vertices as context accumulates. A latent-level sweep shows the residual encodes the **whole** hierarchy, most strongly the locally-predictive deep latents (R^2 up to 0.66) and least strongly the coarse global root. Causal steering on the layer-1 residual confirms the belief is **used**, not merely decodable: steering toward a wrong latent collapses the true next token’s log-probability ($-0.73 \rightarrow -8.5$). We pre-registered our predictions before seeing results; three of four held, and the one miss (root $R^2 < 0.5$) is informative. Finally, a 30-run sweep over 10 random grammars $\times$ 3 replicate seeds shows all four findings — the inverted level gradient, blooming, layer accumulation, and causal steering — replicate across the RHM family with low replicate variance, and that the published **grammar** = 0 numbers sit inside the family distribution. A further 99-run architecture sweep (width, depth, heads, and training budget, one-axis-at-a-time $\times$ 3 grammars $\times$ 3 seeds) shows the phenomenology is capacity-robust down to a low-width floor: decodability rises with width and depth, the inverted gradient holds at every capacity, heads matter least, and decodability is only weakly correlated with how well the model fit the loss.

1. Introduction

Natural data is hierarchical: characters compose into words, words into phrases, phrases into meaning. The Random Hierarchy Model (RHM) of Cagnetta et al. abstracts this into a clean synthetic grammar — a fixed tree in which each high-level symbol expands, via random production rules, into a fixed-length string of lower-level symbols, down to observed leaves. Predicting the next leaf optimally requires inferring the distribution over the **hidden** latent symbols that generated the observed prefix. The optimal next-token predictor **is** the Bayesian belief state over those latents.

This makes the RHM an ideal testbed for **belief geometry**: the hypothesis, developed in the transformer-interpretability literature, that a network trained by next-token prediction comes to represent, in its residual stream, the posterior distribution over the data-generating process’s hidden state — and that this distribution is laid out as a linear (affine) image of the probability simplex. If true, the belief should be (i) linearly decodable, (ii) geometrically organized as a simplex that sharpens with context, (iii) built up additively across layers, and (iv) causally relevant to the model’s output.

The RHM is uniquely suited to test all four claims **exactly**. Unlike natural language, its hidden state has a precise definition and its posterior is computable in closed form by belief propagation on the tree. With small parameters the entire grammar is enumerable, so the probe target is ground truth — not an estimate. This paper trains a tiny transformer on such a grammar and asks whether the residual stream carries the exact posterior, where in the network it lives, what geometry it traces, and whether the model relies on it.

2. The Random Hierarchy Model and exact beliefs

Grammar. We use the defaults of the spec: arity $s = 2$, depth $L = 3$, so each string has $d = s^L = 8$ leaves; per-level vocabulary $v = 8$; and $m = 2$ production rules per parent symbol, chosen uniformly. Rules are drawn once and held fixed, and constrained to be **unambiguous** (the spec’s quick-failure check confirms no two parents share a production). A sample is generated top-down: pick a root class uniformly, recursively expand each symbol by a uniformly chosen rule, and emit the leaf string. With these parameters the grammar admits exactly $v \cdot m^{d-1} = 8 \cdot 2^7 = 1024$ equiprobable distinct trees — small enough to enumerate completely.

The exact frozen grammar used in every result is shown below. Symbols are written as S1, ..., S8 for readability (the saved arrays use zero-based ids); at each level the parent takes one of the two listed child pairs uniformly.

Parent	Top expansion	Middle expansion	Bottom expansion
S1	[S3, S1] or [S4, S8]	[S2, S1] or [S1, S5]	[S4, S1] or [S5, S3]
S2	[S4, S3] or [S1, S3]	[S6, S3] or [S3, S1]	[S6, S7] or [S8, S8]
S3	[S6, S7] or [S5, S6]	[S5, S2] or [S1, S1]	[S3, S4] or [S7, S8]
S4	[S6, S8] or [S7, S7]	[S6, S5] or [S8, S6]	[S6, S5] or [S4, S7]
S5	[S6, S2] or [S1, S5]	[S4, S3] or [S1, S8]	[S1, S5] or [S6, S4]
S6	[S8, S6] or [S2, S7]	[S7, S6] or [S4, S5]	[S8, S4] or [S1, S8]
S7	[S4, S7] or [S1, S1]	[S3, S8] or [S4, S1]	[S4, S6] or [S3, S2]
S8	[S2, S2] or [S5, S7]	[S1, S2] or [S8, S3]	[S5, S7] or [S3, S6]

Table 1: Frozen level-specific grammar sampled once before training. Top, middle, and bottom columns correspond to rules_0, rules_1, and rules_2 in artifacts/grammar.npz.

Exact posterior. Given a leaf prefix of length k , the posterior over any hidden latent (including the root class) is computed by sum-product belief propagation on the tree: upward messages from observed leaves combine through the production rules to give the marginal over each latent. Because the grammar is fully enumerable, we verified the belief-propagation posterior against brute-force enumeration of all consistent trees; the two agree to $< 10^{-6}$ across 10 passing tests, alongside checks that sample length equals s^L and that unambiguity holds. The recorded self-test (a representative string) shows the root posterior sharpening with context — entropy 1.89 nats at $k = 1$ falling to 0.00 (a single consistent root) by $k = 3$ — which previews the blooming geometry below.

3. Model and training

We train a HuggingFace `GPT2LMHeadModel` with 2 layers, `n_embd` = 128, 4 heads, and all dropout disabled, totalling 398,848 parameters. There is no tokenizer: RHM leaf symbols are integers fed directly as `input_ids`, with `vocab_size` = 8 and `n_positions` = 8. We train by next-token prediction on 922 training strings (102 held out) for 4000 steps on Apple MPS.

The Bayes-optimal floor. The value of an enumerable grammar is that we know the best achievable loss. Averaging the exact per-position posterior entropy over the seven predicted positions gives the Bayes-optimal mean next-token cross-entropy: 0.725 nats. The uniform baseline is $\ln 8 = 2.079$. The trained model reaches a final held-out cross-entropy of **0.880 nats** (Figure 1), closing $\frac{2.079-0.880}{2.079-0.725} \approx 89\%$ of the gap. The model has, to a good approximation, learned the true posterior predictor — which is the precondition for asking whether it represents the posterior internally.

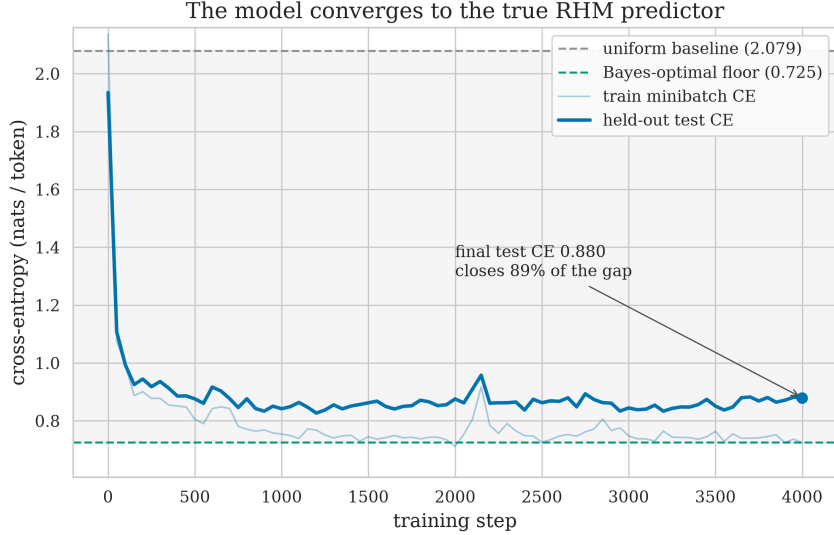


Figure 1: The transformer converges to the true RHM predictor. Held-out next-token cross-entropy (heavy line) falls from the uniform baseline ($\ln 8 = 2.079$, top dashed) toward the Bayes-optimal floor (0.725, bottom dashed) over 4000 training steps, ending at 0.880 nats — closing $\approx 89\%$ of the uniform $\rightarrow$ Bayes gap (shaded). The faint line is the train-minibatch CE. Seeded reproduction of the pinned-arch canonical run (grammar seed 0, $n_{\text{layer}\{=\}}2$, $n_{\text{embd}\{=\}}128$, $n_{\text{head}\{=\}}4$), [results/refrun/](#).

4. Pre-registered prediction

Following the project’s honor-code rule, we committed our predictions to git **before** training any model or computing any probe (`PREREGISTRATION.md`). In brief, we predicted: (1) the root posterior is **linearly decodable** from the residual stream, with $R^2 > 0.5$ at the final layer and a shuffled baseline near 0; (2) the belief **blooms** — early positions cluster near the prior, late positions spread toward simplex vertices, with posterior entropy falling and activation radius growing monotonically with context k ; (3) decodability **accumulates additively across depth**, lowest at the embedding and highest after the last layer; and (4) the representation is **causally used**, so steering the residual along a belief direction shifts the output toward the corresponding latent’s leaves. We also registered alternative outcomes (degenerate collapse, non-linear-only encoding, MAP-only encoding, flat-with-depth) as falsification handles. We report against these predictions in Section 6.

5. Results

5.1. A single linear probe partially decodes the posterior

We fit one global least-squares affine map from the 128-d residual stream to the 8-class exact root posterior, on a probe set of $N = 400$ held-out examples, and score it by R^2 on held-out data. The final-layer probe reaches $R^2 = 0.38$, while the same probe fit to **shuffled** labels scores ≈ 0 (-0.085). So the posterior is genuinely present and linearly accessible above chance — but the probe explains only about a third of the variance: this is a **partial**, noisy recovery, not an exact reconstruction. Projected into a belief-space PCA basis (Figure 2), the probe’s predicted posteriors occupy the same structured region as the exact posteriors and reproduce the same position gradient, but as a diffuse cloud rather than the discrete point set of the ground truth. (The ground-truth panel looks sharper partly because the exact posterior takes few distinct values, so identical points overplot, whereas every probe prediction differs slightly.) The affine correspondence is real, but for the global root it is weak; it strengthens markedly for deeper latents (next subsection) and at later layers.

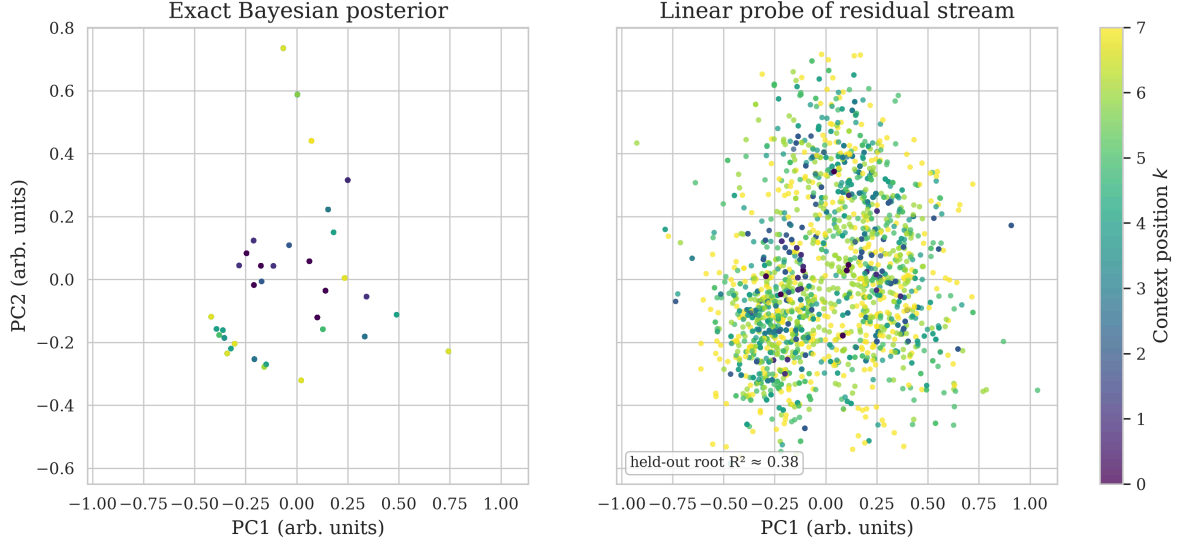


Figure 2: The residual stream is a **partial** affine image of the exact belief simplex. **Left:** PCA(2) of the exact root posteriors (few distinct values, hence sharp overplotted dots); small- k points sit near the prior, large- k points spread toward vertices. **Right:** the linear probe’s predictions in the same basis, colored by context position — the same region and position gradient, but a diffuse cloud: held-out root $R^2 \approx 0.38$, an imperfect recovery, not an exact one.

5.2. Decodability accumulates across depth and context

The residual stream is a running sum of layer contributions, so we ask where the belief is built. Probe R^2 for the root posterior rises monotonically with depth: -0.00 at the embedding (layer 0), 0.15 after layer 1, 0.38 after layer 2, while the shuffled control stays at ≈ 0 throughout (Figure 3, left). Each layer adds belief-relevant signal. Resolving by context position (Figure 3, right), early positions are decodable even at shallow layers, and the final layer pushes near-perfect decodability ($R^2 = 1.0$) across positions 0–3. A few mid-layer cells are strongly negative (the probe underperforms the mean predictor on those positions); the heatmap colors are clipped for legibility, but the cell labels show the raw values.

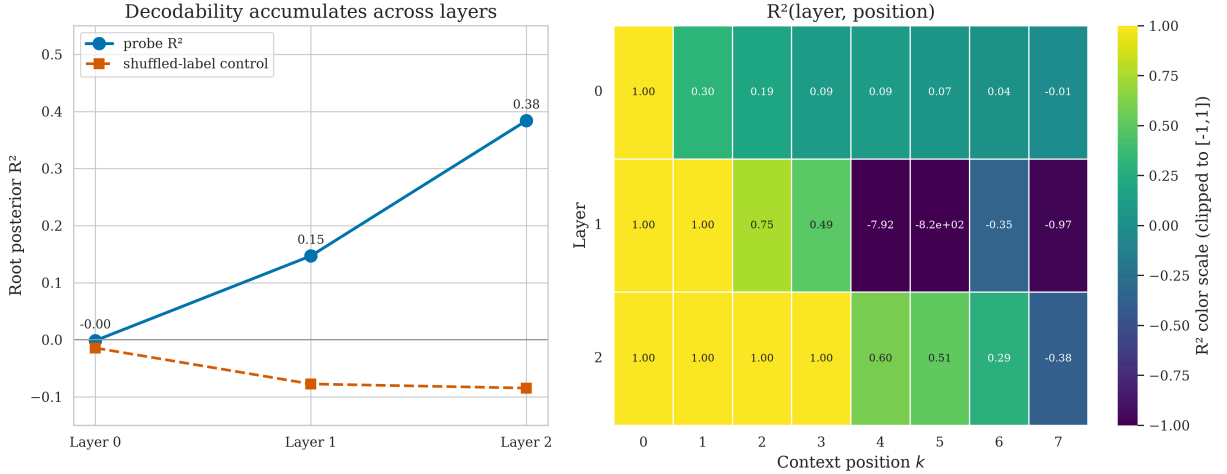


Figure 3: Belief decodability accumulates across depth and context. **Left:** root-posterior R^2 rises layer by layer ($-0.00 \rightarrow 0.15 \rightarrow 0.38$) while a shuffled control stays at ≈ 0 . **Right:** $R^2(\text{layer}, \text{position})$ heatmap; the final layer reaches $R^2 = 1.0$ on the earliest positions. Colors are clipped to $[-1, 1]$, while labels show raw values.

5.3. The whole latent hierarchy is encoded — local latents most strongly

The root is only one of the tree’s hidden latents. Probing the exact posterior over latents at **every** level reveals a clear gradient with some node-level variation (Figure 4): root (L_0) $R^2 = 0.38$; the two level-1 mid latents 0.51 and 0.59 ; and the four deepest level-2 latents 0.61 , 0.66 , 0.50 , and 0.66 . The residual encodes the entire hierarchy, but the **local, near-leaf** latents — those most directly predictive of the next token

— are usually read off more cleanly than the coarse global root. This is our most informative finding: the spec’s primary target (the root) is the **hardest** latent to decode, because it is the most abstract and the least locally predictive.

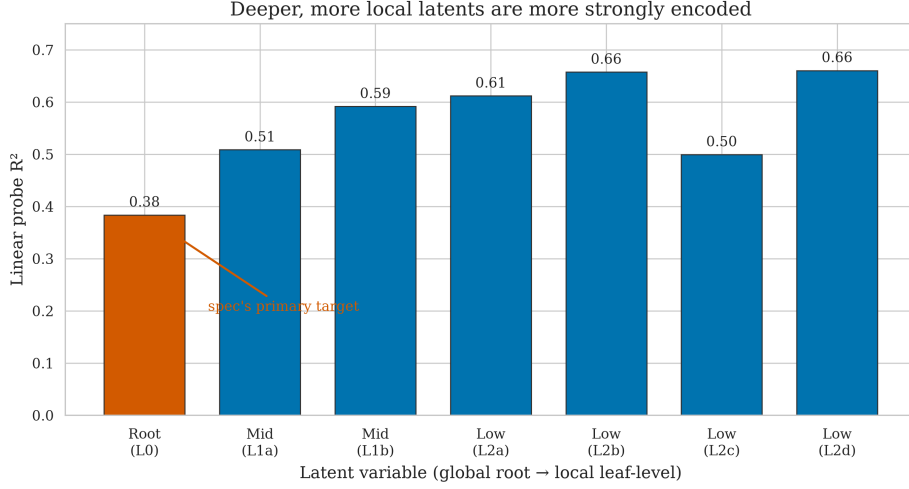


Figure 4: The residual encodes the whole latent hierarchy, deeper/local latents most strongly. Probe R^2 is lowest for the root ($L_0 = 0.38$, the spec’s primary target), higher for level-1 (0.51, 0.59), and generally higher for level-2 (0.61, 0.66, 0.50, 0.66).

5.4. The belief blooms with context

The headline geometry (Figure 5): projecting the linear readout into a belief-space PCA(2) basis, the points form a tight central cluster near the uniform prior when little context is observed and expand outward toward the simplex vertices as context accumulates. The mean radius about the prior anchor grows monotonically from 0.17 at $k = 0$ to ≈ 0.39 at $k = 7$, tracking the true posterior’s own growth (0.17 $\rightarrow$ 0.47); equivalently, the exact posterior entropy falls monotonically with position (1.84 nats at $k = 0$ to 0.00 at $k = 7$). Colored by ground-truth root class, the cloud resolves into separated petals, one per inferred root. Raw residual PCA, dominated by token and position nuisance variance, does **not** bloom — it is the **belief content** of the residual that does.

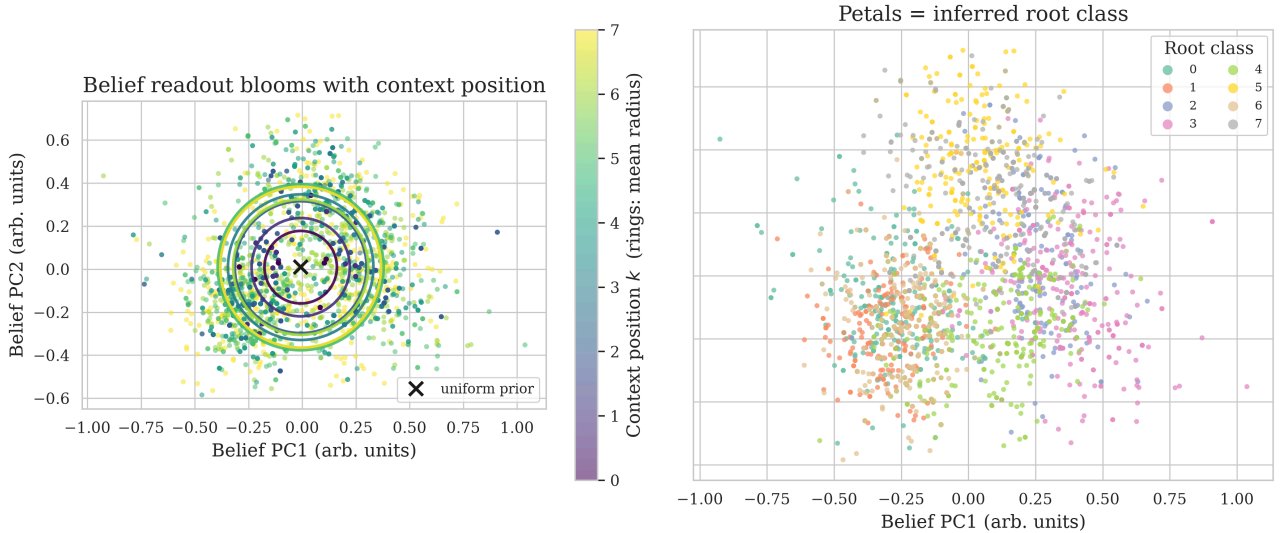


Figure 5: Headline result: the belief read out of the final-layer residual blooms with context. **Left:** points colored by context position k with rings marking each position’s mean radius about the prior ($\times$); the cloud expands outward as context grows (mean radius 0.17 $\rightarrow$ 0.39, tracking the true posterior 0.17 $\rightarrow$ 0.47). **Right:** the same cloud by ground-truth root class resolves into separated petals.

5.5. Causal steering: the belief is used, not just decodable

A representation can be decodable yet causally inert. To distinguish the two we apply a mean-difference patch to the layer-1 residual, pushing it by α times the belief direction toward a chosen latent, and measure

the effect on the next-token distribution (Figure 6). Steering **toward the true** latent leaves predictions untouched (the model already holds that belief): the true token’s log-probability stays at -0.73 for all α . Steering **toward a wrong** latent collapses the mean log-probability of the true next token from -0.73 to -8.5 as α grows, and re-routes probability mass onto the wrong latent’s children (from 0.22 to 0.53). Intervening on the decoded belief axis changes behavior in the predicted direction — the belief state is causally used.

The belief state is causally used, not just decodable

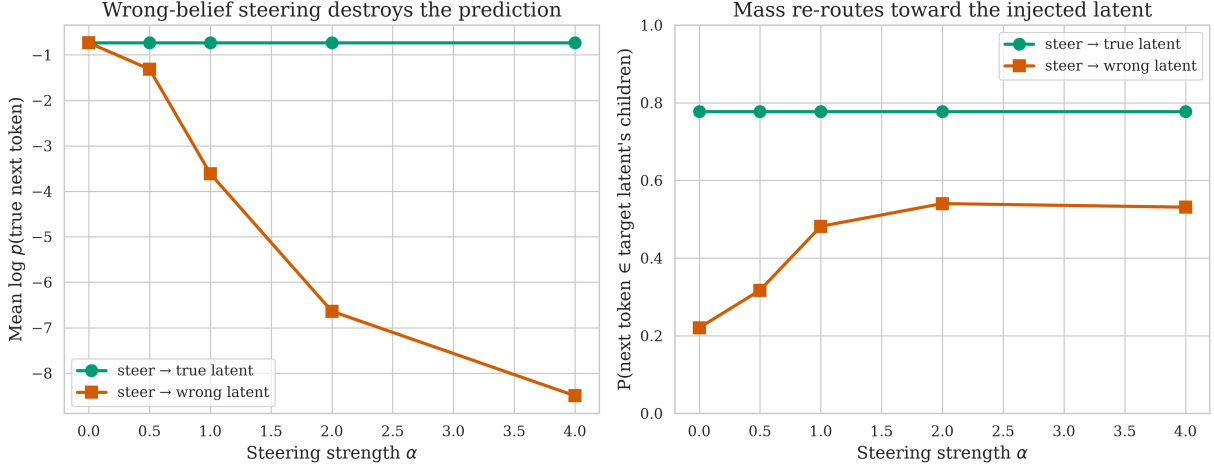


Figure 6: The belief state is causally used. Steering the layer-1 residual toward the **true** latent leaves predictions intact (green); steering toward a **wrong** latent collapses the true next token’s log-probability (left, orange: $-0.73 \rightarrow -8.5$) and re-routes mass onto the wrong latent’s children (right, orange: $0.22 \rightarrow 0.53$).

6. Pre-registration scorecard

We grade each pre-registered prediction honestly against the outcome.

Prediction	Outcome	Verdict
Linear decodability, root $R^2 > 0.5$, shuffled ≈ 0	Root $R^2 = 0.38$ (shuffled ≈ 0). Decodable and well above baseline, but below the 0.5 threshold for the root; mid/deep latents do exceed 0.5 (up to 0.66).	Partial
Blooming with context (radius $\uparrow$, entropy $\downarrow$, monotone)	Mean radius $0.17 \rightarrow 0.39$ monotone; posterior entropy $1.84 \rightarrow 0.00$ monotone.	Hit
Monotone layer-wise accumulation of R^2	$-0.00 \rightarrow 0.15 \rightarrow 0.38$ across the three residual indices.	Hit
Causal usage via steering (ordered effect)	Steering $\rightarrow$ wrong: log p collapses $-0.73 \rightarrow -8.5$, monotone in α ; target-child mass $0.22 \rightarrow 0.53$.	Hit

Three of four predictions held cleanly. The single miss is the root R^2 , which came in at 0.38 rather than the predicted > 0.5 . We take this as a genuine and informative negative: the prediction was correct in **form** (linear, above baseline, growing with depth and context) but our magnitude estimate for the **root specifically** was optimistic. The latent-level sweep — which we did not pre-specify — explains why: the root is the coarsest, least locally-predictive latent, and the threshold we predicted is in fact met by every deeper latent in the hierarchy. None of the pre-registered failure modes (degenerate collapse, non-linear-

only encoding, MAP-only encoding, flat-with-depth) materialized; in particular MAP root-class accuracy is only 0.47 while full-simplex R^2 is positive and graded, ruling out a MAP-only representation.

7. Generalization across grammars

The results above come from a single grammar (`grammar = 0`) and a single training run. To test whether they are properties of the RHM **family** under our fixed constraints ($s = 2$, $L = 3$, $v = 8$, $m = 2$, uniform unambiguous rules) rather than artifacts of one rule draw, we sweep the **content of the rule table** across ten independently sampled grammars (`Grammar.random(seed = 0.9)`) with three replicate training seeds each — 30 runs at the pinned architecture ($n_{\text{layer}} = 2$, $n_{\text{embd}} = 128$, $n_{\text{head}} = 4$) and full 4000-step budget. A single master seed controls model initialisation, training-data sampling, probe sampling, and the probe split, so replicates measure end-to-end sampling variance. Each run writes a deterministic, self-contained directory (`results/sweep/g{NN}_L2_d128_h4_s{S}/`) holding its grammar, a `config.json` manifest (resolved config, git SHA, timestamp), and per-run metrics — recoverable for this paper independent of any experiment-tracker. Published `grammar = 0` is the first cell of the sweep, so its number is locatable in the distribution as a built-in consistency check.

The ten grammars are not isomorphic. Each grammar’s rule table is an adjacency structure (which parent expands to which ordered children, per level); two grammars are the same up to renaming iff a per-level symbol permutation makes those tables coincide. An exact test for such a permutation finds all $\binom{10}{2} = 45$ pairs distinct (`grammar_iso.py`, checkpointed in `tests/`).

All grammars train to near-Bayes. Every run closes 0.82–0.93 of the uniform→Bayes loss gap (mean 0.89 ± 0.03), matching the pass-1 value (0.88); no run failed to train, so we report all 30 with no convergence filtering (Table 4). Loss-gap-closed is shown as a sanity covariate, not used to gate any run.

The inverted strength gradient replicates as a distribution (Figure 7). Pooling probe R^2 by tree level, the root (L_0) averages 0.35 ± 0.07 (range 0.24–0.48), well below the mid (L_1 , mean 0.50) and leaf-parent (L_2 , mean 0.49) latents. The root is the weakest-decoded level across the **whole family**, not just in the published draw: the coarse global latent is the hardest to read off, while the locally-predictive deeper latents are encoded more strongly. The pass-1 root $R^2 = 0.38$ sits comfortably inside the replicate band, confirming it as a typical rather than cherry-picked draw. The deepest level shows the widest spread (one grammar dips slightly negative), as expected — node-level rule structure matters most for the most local latents.

Blooming and belief-sharpening hold for every grammar (Figure 8). The exact posterior entropy falls monotonically with context position k in all ten grammars, and the belief readout’s radius grows with context — the blooming geometry is a family-wide property, not a feature of one rule table.

Causal steering replicates with a stable effect size (Figure 9). Steering the layer-1 residual toward a wrong latent collapses the true next token’s mean log-probability in every run, by 6.9 ± 0.5 nats from $\alpha = 0$ to $\alpha = 4$. The belief is causally used across the whole grammar family, with low run-to-run variance. In short, all four pass-1 findings — inverted level gradient, blooming, layer-wise accumulation (recomputed in every run’s per-layer probe), and causal steering — replicate across the grammar family and are stable across replicate sampling. The headline phenomenology is a property of the RHM under these constraints, not of one rule draw.

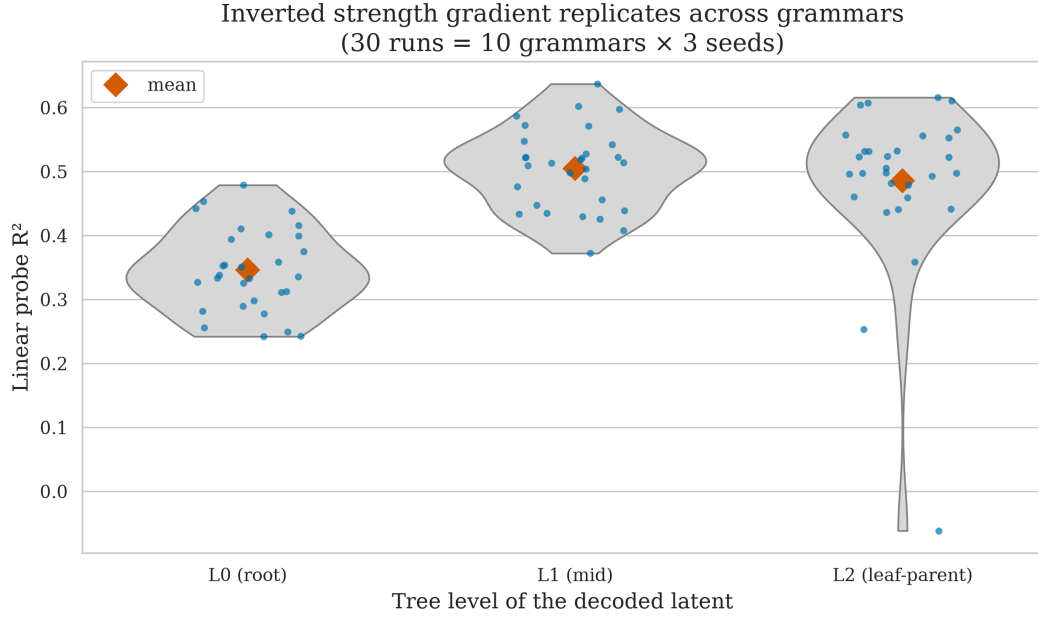


Figure 7: The inverted strength gradient replicates across the grammar family. Each point is one of 30 runs (10 grammars $\times$ 3 seeds); violins show the per-level distribution of probe R^2 , diamonds the means. The root (L_0) is the weakest-decoded level (0.35 ± 0.07), below the mid (L_1) and leaf-parent (L_2) latents (≈ 0.49 – 0.50).

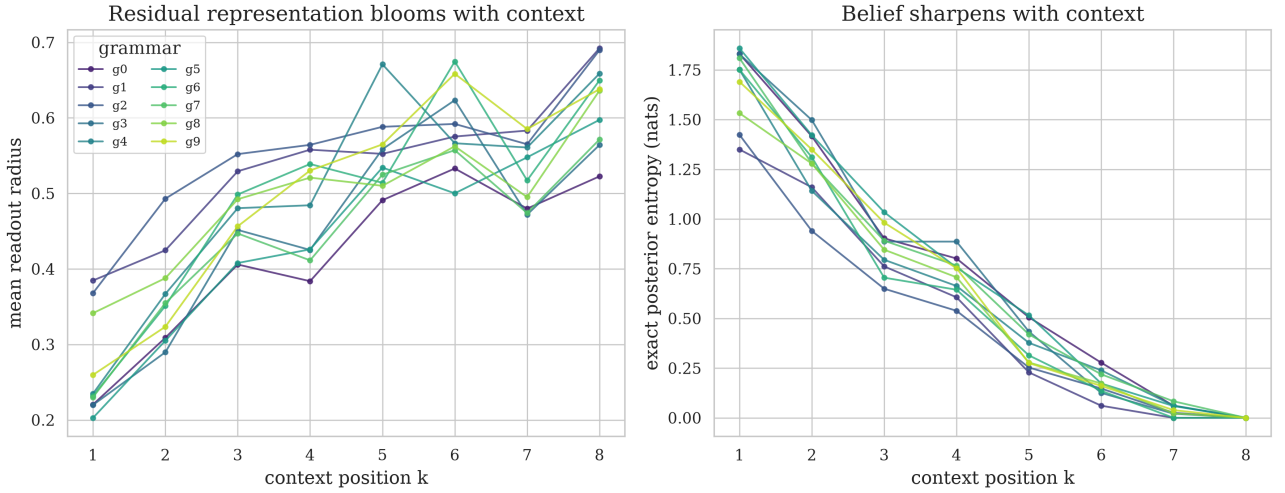


Figure 8: Blooming is family-wide. **Left:** the belief readout’s mean radius grows with context position k (one line per grammar, averaged over seeds). **Right:** the exact posterior entropy falls monotonically with k for every grammar.

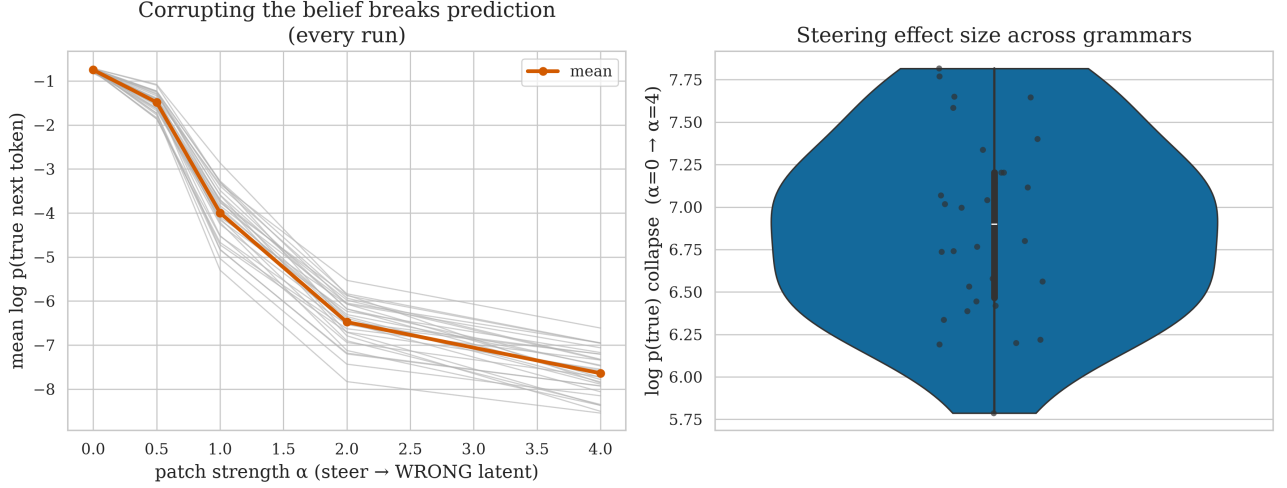


Figure 9: Causal steering replicates with a stable effect size. **Left:** steering the layer-1 residual toward a **wrong** latent collapses the true next token’s log-probability in every run (grey lines), mean in orange. **Right:** the distribution of the collapse magnitude ($\alpha=\{0 \rightarrow \alpha=\{4\}$) across all 30 runs: 6.9 ± 0.5 nats.

8. Architecture dependence

The grammar sweep fixes the architecture and varies the data; this section does the opposite. We hold the RHM constraints fixed ($s = 2, L = 3, v = 8, m = 2$) and vary the transformer’s capacity one axis at a time (OAT) around the published pass-1 baseline ($n_{\text{layer}} = 2, n_{\text{embd}} = 128, n_{\text{head}} = 4, 4000$ steps): width $n_{\text{embd}} \in \{16, 64, 128, 256\}$, depth $n_{\text{layer}} \in \{1, 2, 3, 4\}$, heads $n_{\text{head}} \in \{1, 2, 4, 8\}$, and training budget steps $\in \{4000, 16000\}$. Each axis varies independently with the other three pinned at baseline, so the eleven distinct configs share the baseline as their common center. Every config is run across `grammar` $\in \{0, 1, 2\}$ and master seed $\in \{0, 1, 2\}$ — **99 runs total** (run_arch.py, writing steps-disambiguated dirs `results/arch/g{NN}_L{n}_d{d}_h{h}_t{steps}_s{S}/`). No convergence filtering: small models are **expected** to underfit, and we carry `loss_gap_closed` as a covariate rather than a gate. This builds on the pass-1 pre-registration (PREREGISTRATION.md); the expectations below are stated for framing, not pre-registered.

Capacity lifts decodability, with a low-width floor (Figure 10). Width is the strongest lever: across $n_{\text{embd}} = 16 \rightarrow 256$ the root probe R^2 climbs monotonically $0.07 \rightarrow 0.21 \rightarrow 0.35 \rightarrow 0.44$, with the mid ($0.12 \rightarrow 0.57$) and leaf-parent ($0.15 \rightarrow 0.59$) levels rising in parallel; at $n_{\text{embd}} = 16$ the root nearly collapses ($R^2 = 0.07$, barely above zero) — a genuine minimum-capacity floor below which the coarse latent is no longer linearly present. Depth lifts every level too: across $n_{\text{layer}} = 1 \rightarrow 4$ the root goes $0.22 \rightarrow 0.42$, the mid $0.32 \rightarrow 0.54$, and the leaf-parent $0.34 \rightarrow 0.48$. Heads are the weakest axis, as anticipated: from 1 to 8 heads the root moves only $0.25 \rightarrow 0.36$, the mid $0.37 \rightarrow 0.48$, and the leaf-parent level is essentially flat ($0.42 \rightarrow 0.44$). One framing expectation does **not** survive: we guessed the root would be information-limited and gain little from extra capacity, but width and depth lift it substantially — the root is the weakest latent at **every** capacity, yet it is far from saturated.

The inverted gradient is capacity-robust. At all eleven configs the root (L_0) is the weakest-decoded level, below the mid (L_1) and leaf-parent (L_2) latents — the pass-1/2 inverted strength gradient is not an artifact of the baseline size. It holds at the smallest width (where everything is low) and the largest (where everything is high). The effect is a **two-tier split**, not a monotone ladder: the root sits clearly below the local latents (mean $R^2 = 0.31$ vs $0.42/0.41$; L_0 is the strict minimum in 88 of 99 runs, with an $L_1 - L_0$ gap of 0.11 ± 0.06), while the mid and leaf-parent levels are statistically indistinguishable from each other ($L_1 < L_2$ in only 68 of 99 runs). “Inverted” refers to the **direction** — the coarse global root, the naive prediction’s strongest target, is in fact the hardest to read off — not to a clean $L_0 < L_1 < L_2$ ordering.

Depth stretches the build-up and lifts the readout (Figure 11). Plotting root R^2 against normalized residual depth (residual index over n_{layer}), every model rises from ≈ 0.01 at the embedding to its final-layer readout, and deeper models both stretch the accumulation curve and reach a higher endpoint: final-layer

root R^2 is 0.22, 0.35, 0.39, 0.42 for $n_{\text{layer}} = 1, 2, 3, 4$. Belief assembly is not a two-layer accident — it uses whatever depth it is given.

Longer training helps the root more than the leaves. Quadrupling the budget (4000 $\rightarrow$ 16000 steps) lifts the root by +0.05 (0.35 $\rightarrow$ 0.41) and the mid level by +0.05 (0.48 $\rightarrow$ 0.53), but the leaf-parent level by only +0.02 (0.44 $\rightarrow$ 0.47). The coarse, globally-determined latent is the slowest to be linearized, consistent with it being optimization-limited rather than already-saturated.

Decodability decouples from loss fit (Figure 12). Across all 99 runs the correlation between `loss_gap_closed` and probe R^2 is modest — 0.46 for the root, 0.42 for the mid, 0.17 for the leaf-parent — and the scatter is wide: models that close the same fraction of the loss gap span a large range of decodability (root R^2 from below 0.1 to above 0.6 at `loss_gap_closed` $\approx$ 0.9). Linear belief decodability is therefore **not** a restatement of how well the model fit the next-token loss; a model can reach near-Bayes loss without linearly representing the coarse belief, and vice versa.

Marginal capacity effects on belief decodability (OAT around L2·d128·h4·t4000; error bars = SEM over 3 grammars $\times$ 3 seeds)

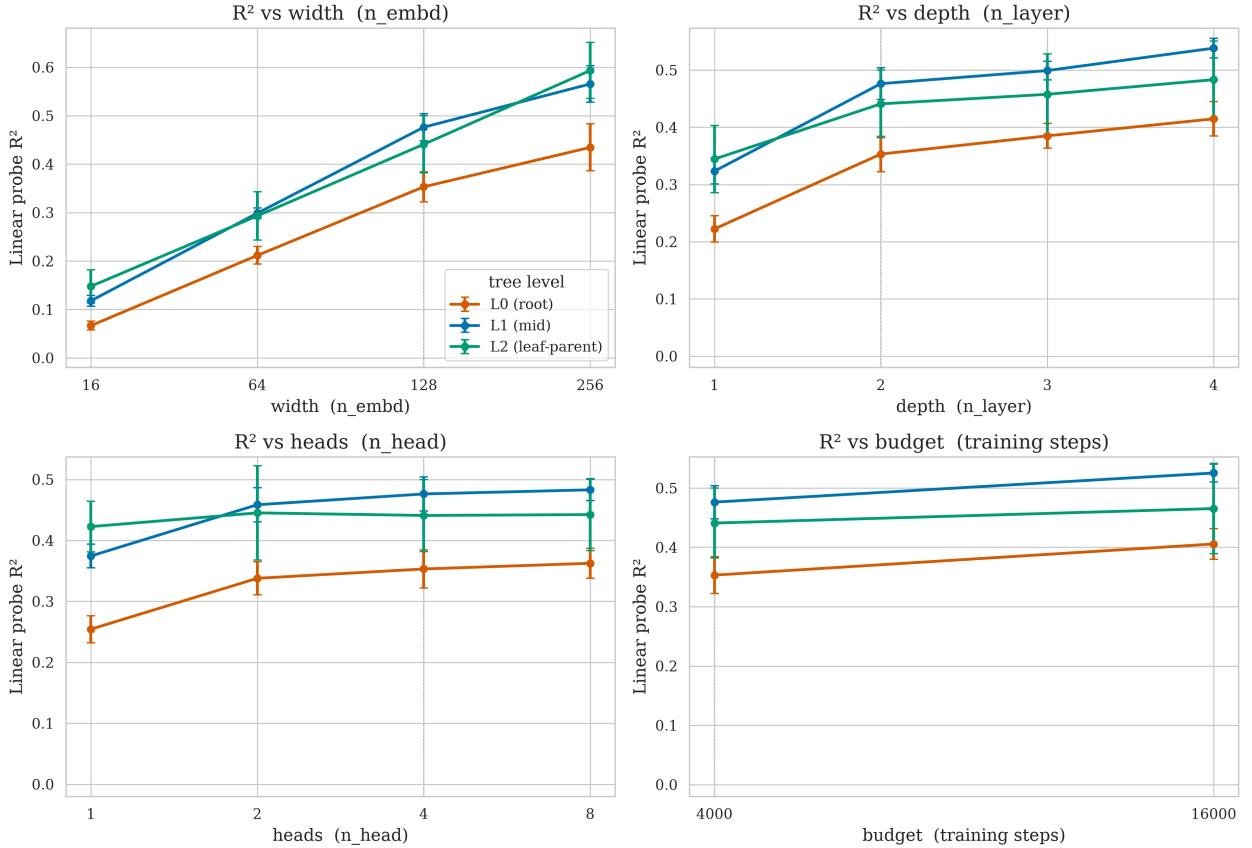


Figure 10: Marginal capacity effects (OAT). Each panel varies one axis with the other three at baseline; points are root (L_0), mid (L_1), and leaf-parent (L_2) probe R^2 , error bars are SEM over 3 grammars $\times$ 3 seeds, the dotted line marks the shared baseline. Width and depth lift all levels (root collapses at $n_{\text{embd}} = 16$); heads matter least; the root is the lowest level in every panel.

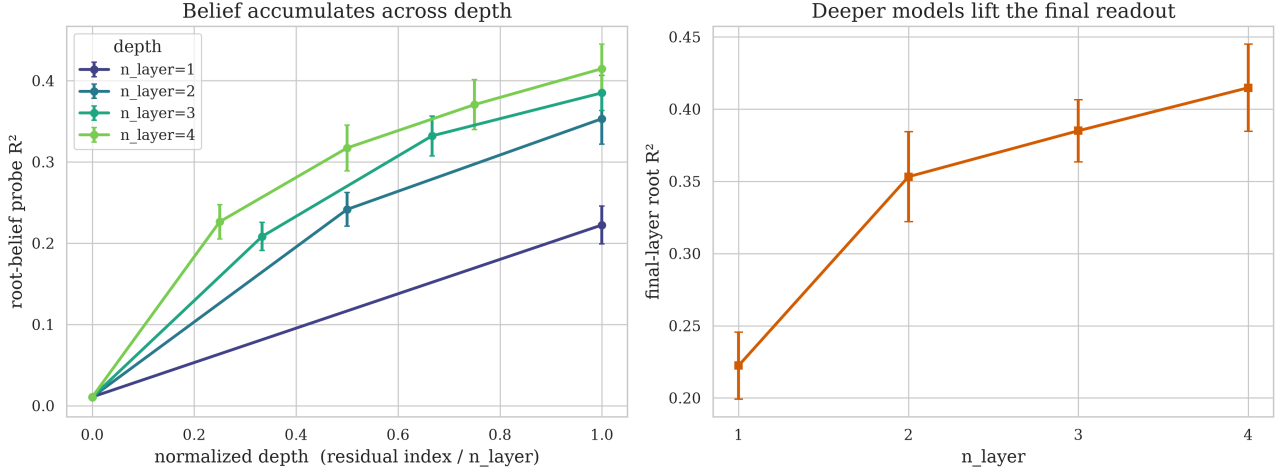


Figure 11: Depth and accumulation. **Left:** root-belief probe R^2 vs normalized residual depth, one line per n_{layer} (mean $\pm$ SEM over grammars $\times$ seeds); every model accumulates from the embedding to its readout, and deeper models reach higher. **Right:** final-layer root R^2 rises monotonically with depth.

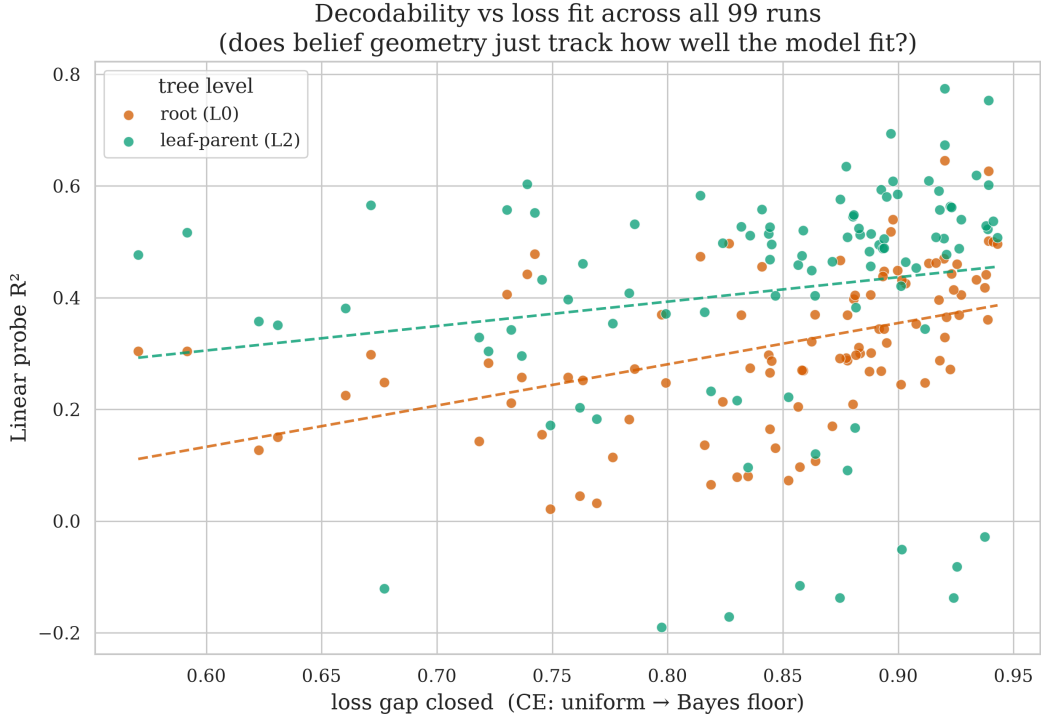


Figure 12: Decodability vs loss fit, all 99 runs. Root (L_0) and leaf-parent (L_2) probe R^2 against `loss_gap_closed` (fraction of the uniform $\rightarrow$ Bayes CE gap closed). Dashed lines are least-squares fits; the wide vertical spread at fixed loss fit shows decodability is not determined by how well the model fit the loss.

9. Going deeper: $L = 4$

The $L = 3$ tree is shallow enough that the root collapses to certainty within a few tokens, compressing the window over which its belief could be seen to bloom. The pass-1 paper named $L = 4$ as the follow-up that should give the **root** more dynamic range. This pass runs the full replication one tree level deeper — depth $L = 4$, so context length $d = s^L = 16$ (double pass-1’s 8) and a **fourth** latent level: root L_0 (1 node), L_1 (2), L_2 (4), and the leaf-parents L_3 (8) above the 16 observed leaves. Everything else stays vanilla ($s = 2, v = 8, m = 2$, uniform, unambiguous). We sweep ten independently sampled, **pairwise non-isomorphic** $L = 4$ grammars (re-verified at $L = 4$, `grammar_iso.py`) $\times$ three master seeds $\times n_{\text{layer}} \in \{2, 3\}$ — **60 runs** (`run_depth4.py`, writing `results/depth4/g{NN}_L{n}_d128_h4_t4000_s{S}/`, a separate root so the depth-agnostic dir names never collide with pass-2/3). Every run closes 0.96–0.99 of the uniform $\rightarrow$ Bayes loss gap (mean 0.99); none failed to train, so all 60 are reported with no filtering (Table 6).

Critically, the three predictions below were committed (PREREGISTRATION_L4.md, git SHA) **before** any $L = 4$ result was computed; we grade them in Table 2.

All four pass-1 findings replicate at $L = 4$. Linear decodability holds (every level is read off well above the shuffled baseline of ≈ -0.04); the belief sharpens with context (exact root posterior entropy falls $1.86 \rightarrow 0.00$ monotonically over the 16 positions, Figure 14); belief **accumulates across layers** (root probe R^2 rises $-0.01 \rightarrow 0.07 \rightarrow 0.19$ for $n_{\text{layer}} = 2$ and $-0.01 \rightarrow 0.06 \rightarrow 0.17 \rightarrow 0.28$ for $n_{\text{layer}} = 3$, the deeper model reaching a higher readout, Figure 15); and causal steering still collapses the true next token’s log-probability by 6.6 ± 0.5 nats toward a wrong latent (Figure 16). The headline phenomenology survives the depth increase intact.

Prediction 1 (root weaker at $L = 4$) — hit. On the arch-matched cells ($n_{\text{layer}} = 2$, same $n_{\text{embd}} = 128, n_{\text{head}} = 4$, 4000 steps), the root probe R^2 falls to 0.19, well below the $L = 3$ $n_{\text{layer}} = 2$ family mean of 0.35. Adding a third layer lifts it to 0.28, still below the $L = 3$ value. Pushing the root four expansion levels from the leaves does make the coarse global latent harder to assemble linearly, as predicted.

Prediction 2 (root will not bloom) — miss. The extra depth **does** let the root bloom: the exact root posterior entropy falls cleanly and monotonically from 1.86 nats to 0 across the 16 positions, and the belief readout radius grows overall from 0.18 to 0.50 (Figure 14). The growth is real but noisier than the local latents’ — the radius wobbles in the back half rather than rising strictly monotonically — so the bloom is visible but not as crisp as the entropy fall. The pre-registered intuition that the root would resolve abruptly or stay diffuse was wrong; the longer ambiguity window is exactly what the $L = 3$ paper hoped it would be for the root.

Prediction 3 (clean monotone ladder $L_0 < L_1 < L_2 < L_3$) — miss. The per-level gradient is **not** a clean ramp (Figure 13). Family-mean probe R^2 is $L_0 = 0.24, L_1 = 0.37, L_2 = 0.36, L_3 = 0.30$: it **rises** from the root to the mid-levels and then **falls** back toward the leaf-parents — an inverted-U, not an increasing ladder. Only 6 of 60 runs show the strict $L_0 < L_1 < L_2 < L_3$ ordering. What **does** hold is “root weakest”: L_0 is the lowest-decoded level in 50 of 60 runs. The new $L = 4$ wrinkle is that the leaf-parent level L_3 — tied for strongest at $L = 3$ — is no longer the strongest at $L = 4$; the **mid** latents (L_1, L_2) are. With eight leaf-parents each governing only two of sixteen leaves, the most local latents are spread thin, and the position-pooled probe reads the mid-scale latents most cleanly. The “inverted” direction (coarse root hardest) persists; the monotone-ramp sharpening we predicted does not.

$L = 4$ prediction	Outcome	Verdict
1. Root R^2 lower than at $L = 3$ (arch-matched $n_{\text{layer}} = 2$, vs ≈ 0.35)	Root $R^2 = 0.19$ at $n_{\text{layer}} = 2$ (0.28 at $n_{\text{layer}} = 3$), both below the $L = 3$ family mean of 0.35.	Hit
2. The root will not visibly bloom over the longer window	Exact root entropy falls cleanly $1.86 \rightarrow 0.00$; readout radius grows $0.18 \rightarrow 0.50$ (non-monotone). The root does bloom.	Miss
3. Clean monotone ladder $R^2(L_0) < R^2(L_1) < R^2(L_2) < R^2(L_3)$	Means $\frac{0.24}{0.36}$ — inverted-U, mid-levels strongest; strict ladder in only $\frac{6}{60}$ runs. Root still weakest in $\frac{50}{60}$.	Miss

Table 2: Scored $L = 4$ pre-registered predictions (PREREGISTRATION_L4.md, committed before any $L = 4$ result). One of three held.

One of three predictions held. The two misses are informative, not embarrassing: the root **did** gain dynamic range from the deeper tree (contra prediction 2 — the spec’s original motivation for $L = 4$ was right and our pre-registered skepticism was wrong), and the level gradient sharpened into an **inverted-U** rather than the monotone ramp we guessed (contra prediction 3), while the core “coarse root is hardest to decode” direction held. The exact, falsifiable setup let the data correct two genuine priors.

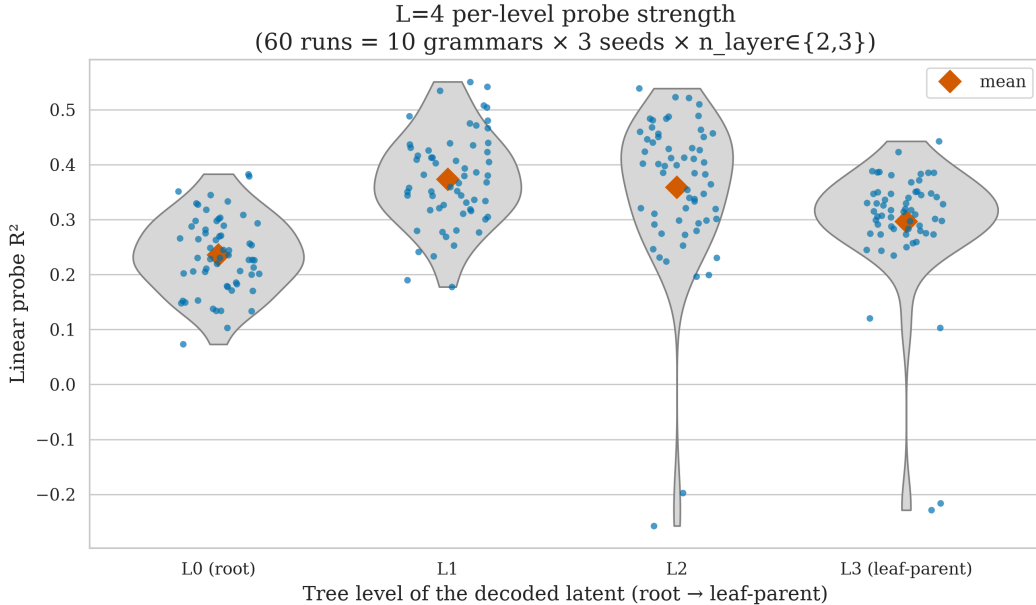


Figure 13: $L = 4$ per-level probe strength across the family. Each point is one of 60 runs (10 grammars $\times$ 3 seeds $\times$ $n_{\text{layer}} \in \{2, 3\}$); violins show the per-level R^2 distribution, diamonds the means. The gradient is an **inverted-U** — the mid-levels (L_1, L_2) decode strongest, the root (L_0) weakest, the leaf-parents (L_3) intermediate — not the pre-registered monotone ladder.

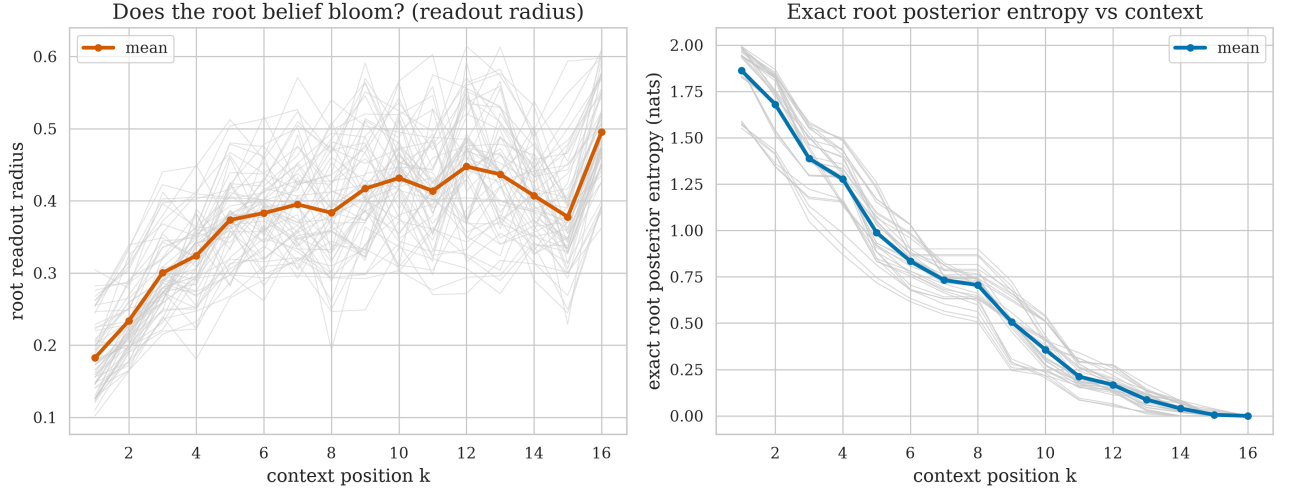


Figure 14: The root **does** bloom at $L = 4$. **Left:** the root belief readout radius grows with context position k over the 16-position window (grey: 60 runs; orange: mean), though non-monotonically in the back half. **Right:** the exact root posterior entropy falls cleanly and monotonically from 1.86 nats to 0.

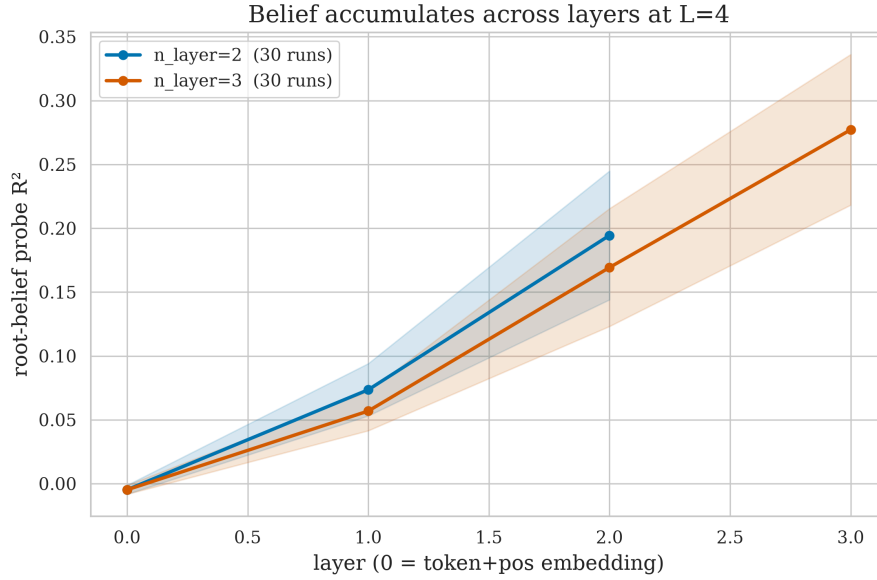


Figure 15: Belief accumulates across layers at $L = 4$. Root-belief probe R^2 vs residual index, one line per n_{layer} (mean $\pm$ SD over 30 runs each). Both archs rise monotonically from the embedding to the readout; the 3-layer model reaches a higher endpoint (0.28 vs 0.19).

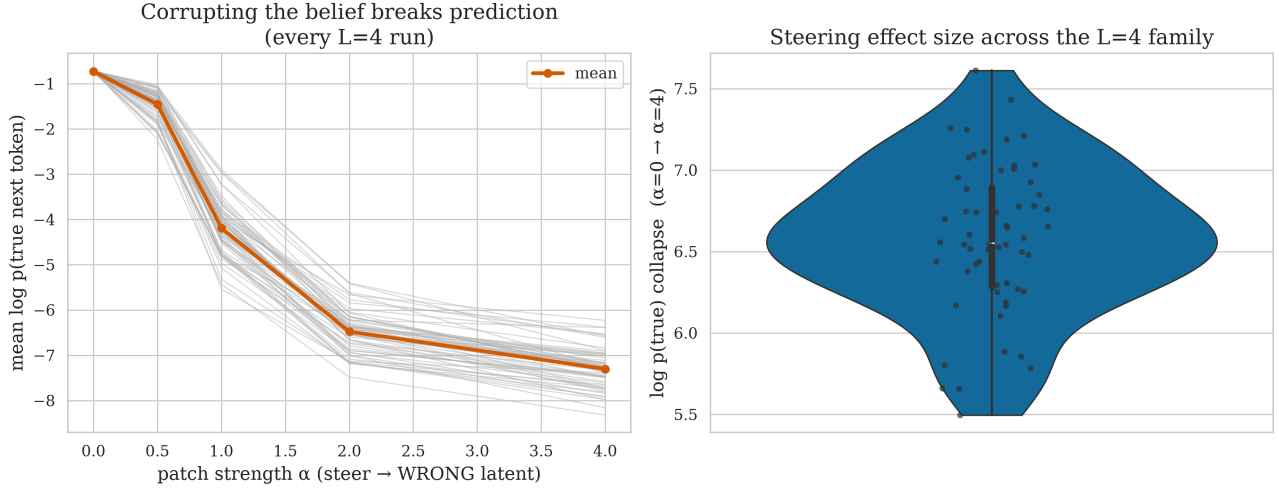


Figure 16: Causal steering replicates at $L = 4$. **Left:** steering the layer-1 residual toward a **wrong** latent collapses the true next token’s log-probability in every run (grey), mean in orange. **Right:** the collapse magnitude ($\alpha=\{0\} \rightarrow \alpha=\{4\}$) across the 60 runs: 6.6 ± 0.5 nats.

10. Engineering non-collapsing belief geometry

Everything so far lives on a tree whose belief, given the full leaf string, collapses to **certainty**: with uniform unambiguous rules the eight leaves invert level-by-level to exactly one root, so the $k = 8$ posterior is a delta and the belief path runs from the prior straight to a simplex vertex. Simplex’s “interesting” HMMs behave oppositely — their belief mixes slowly and never reaches a vertex — but the RHM has no recurrence, so “slow forgetting” has no analog. The native analog of “the state can’t be restored” is a **non-invertible observation channel**: a grammar where even the full leaf string leaves the root uncertain, so the reachable-belief set is a non-trivial attractor in the simplex rather than a path to a vertex. This pass asks what it takes to engineer that, and whether the linear probe still recovers it.

We add two per-level knobs to the grammar (`rh.m.py`, both reducing exactly to the canonical RHM at their off setting, verified byte-identical). **Ambiguity** ρ shares a fraction of the $v \cdot m$ rule entries’ child-tuples **across parents** (many-to-one leaf $\rightarrow$ root structure); $\rho \in \{0, 0.3, 0.6\}$. **Skew** draws each parent’s rule-choice probabilities non-uniform (Dirichlet concentration: *none, mid, high*). Both belief propagation and the brute-force reference were generalized to **weighted** sum-product using the same probabilities the sampler uses; the $\text{BP} \leftrightarrow$ brute-force agreement holds to 2.5×10^{-16} under both knobs simultaneously. We sweep the full 3×3 grid $\times 3$ rule-table draws — **27 runs** at the pinned published arch (`run_noncollapse.py`, writing `results/noncollapse/`), with the (*none, none*) cell reproducing the pass-1 baseline.

The predictions below were committed (PRREGISTRATION-`noncollapse.md`, git SHA) **before** any result of this study was computed; we grade them in Table 3.

Ambiguity engineers non-collapse; skew alone does not (the headline). Mean exact $k = 8$ root-posterior entropy is **exactly 0** at $\rho = 0$ — for **every** skew level — and rises monotonically with ρ : $0.00 \rightarrow 0.71 \rightarrow 1.38$ nats at *skew=none* (Figure 17, left). Skewing the rule-choice probabilities bends the **intermediate** trajectory (the back-half entropy sits higher) but the full string still inverts uniquely, so the endpoint stays a delta: the $\rho = 0$ column is a flat zero across *none/mid/high*. The full entropy-vs- k trajectory (Figure 17, right) makes the phenomenon legible — at $\rho = 0$ the belief collapses cleanly to 0 by $k = 8$ (the bloom-to-vertex), while at $\rho = 0.3$ and $\rho = 0.6$ it **plateaus** at a positive floor: a genuine non-collapsing attractor. Higher skew **lowers** the plateau (it concentrates the surviving posterior) but never removes it. This isolates ambiguity, a property of the decoding channel, as the only knob that manufactures irreducible uncertainty — exactly as pre-registered.

The linear probe survives — and, surprisingly, sharpens. At every cell of the grid the residual stream still linearly encodes the (now spread) posterior well above the shuffled-label baseline of ≈ -0.07 (Figure 18). We pre-registered that the probe would **degrade** as ρ rose, the spread posterior being a harder regression target. The data overturn this: root probe R^2 **rises** with ambiguity, $0.36 \rightarrow 0.42 \rightarrow 0.53$ at *skew=none*, and the mid- and low-level probes rise even more steeply ($L_1 : 0.46 \rightarrow 0.71$; $L_2 : 0.35 \rightarrow$

0.82). A delta-collapsed posterior is a near-constant target once the context pins it, starved of variance; a non-collapsing posterior is a richer, higher-variance, more linearly-structured signal, so linear decodability **improves**. The attractor comparison (Figure 19) shows the probe readout tracing the same structured interior cloud as the exact posterior at $\rho = 0.6$, versus the vertex-bound bloom at $\rho = 0$. The non-collapsing geometry is not just achievable — it is **more** linearly recoverable than the collapsing one.

Non-collapse prediction	Outcome	Verdict
1. $\rho > 0 \implies k = 8$ root entropy > 0 , monotone in ρ ; reachable set gains spread	skew=none $k = 8$ entropy $0.00 \rightarrow 0.71 \rightarrow 1.38$ nats; strictly increasing; belief plateaus instead of collapsing.	Hit
2. skew >0 , rho=0 still collapses ($k = 8$ entropy ≈ 0)	$k = 8$ entropy is exactly 0 at $\rho = 0$ for none/mid/high. Skew reshapes the path, adds no endpoint uncertainty.	Hit
3. Per-level probe R^2 degrades with ρ (but stays above shuffled)	Stays above shuffled everywhere ($\checkmark$), but R^2 rises with ρ ($0.36 \rightarrow 0.53$ at root), not degrades — the richer target is easier to decode.	Partial

Table 3: Scored non-collapse pre-registered predictions (PREREGISTRATION-noncollapse.md, committed before any result of this study). Two of three held; the third’s “stays decodable” clause held while its “degrades” clause was overturned.

Two of three predictions held outright; the third split — the probe stays robust (as predicted) but the direction of the effect inverts (decodability **improves** with ambiguity), a falsifiable prior corrected by the data. As a covariate, the Bayes next-token floor **rises** with ρ at fixed skew (more genuine ambiguity) but **falls** sharply with skew (near-deterministic rule choice makes the next token predictable): e.g. skew=none floor ≈ 0.72 – 0.78 nats versus skew=high floor ≈ 0.16 – 0.40 . The fraction of the uniform→Bayes gap the tiny model closes stays ≈ 0.93 – 0.96 at skew $\leq$ mid, dipping to ≈ 0.5 – 0.7 at skew=high — an optimization effect (the sharp low floor is harder to reach in 4000 steps), not a belief-geometry effect, since the probe targets the **exact** posterior. We note one numerical caveat: convex-hull area and effective dimensionality of the reachable set do **not** cleanly separate the regimes, because a collapsed posterior places its deltas at **distinct** simplex vertices — themselves a spread set; root posterior **entropy** is the clean non-collapse discriminator and is what the headline curve reports.

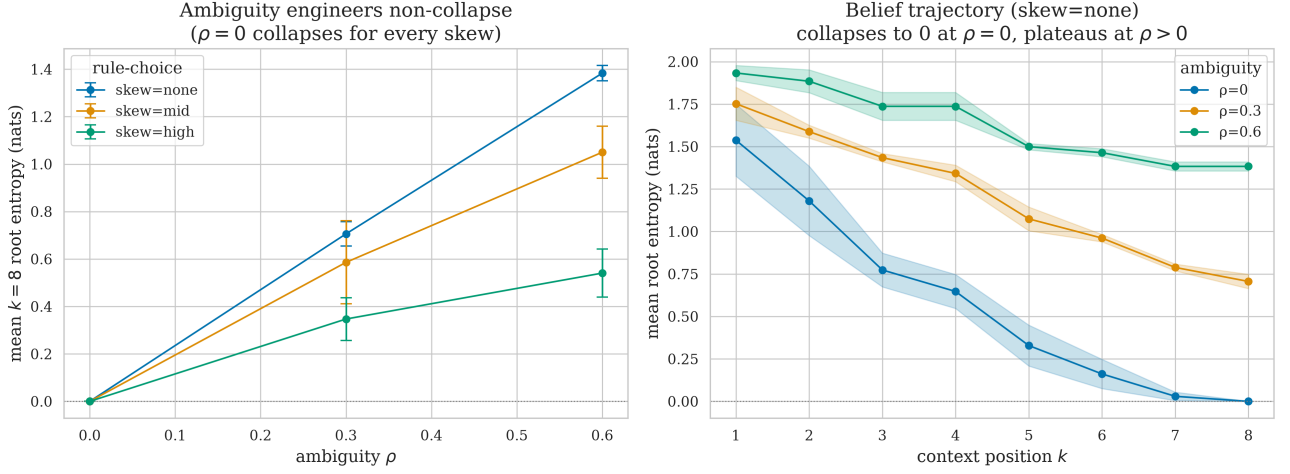


Figure 17: Ambiguity, not skew, engineers non-collapse. **Left:** mean exact $k = 8$ root posterior entropy vs ambiguity ρ , one line per skew (error bars over 3 draws). At $\rho = 0$ entropy is exactly 0 for **every** skew — skew alone does not prevent collapse — and it rises monotonically with ρ . **Right:** the full entropy-vs-context trajectory at skew=none; at $\rho = 0$ the belief collapses to 0 by $k = 8$, at $\rho > 0$ it plateaus at a positive floor — a non-collapsing attractor.

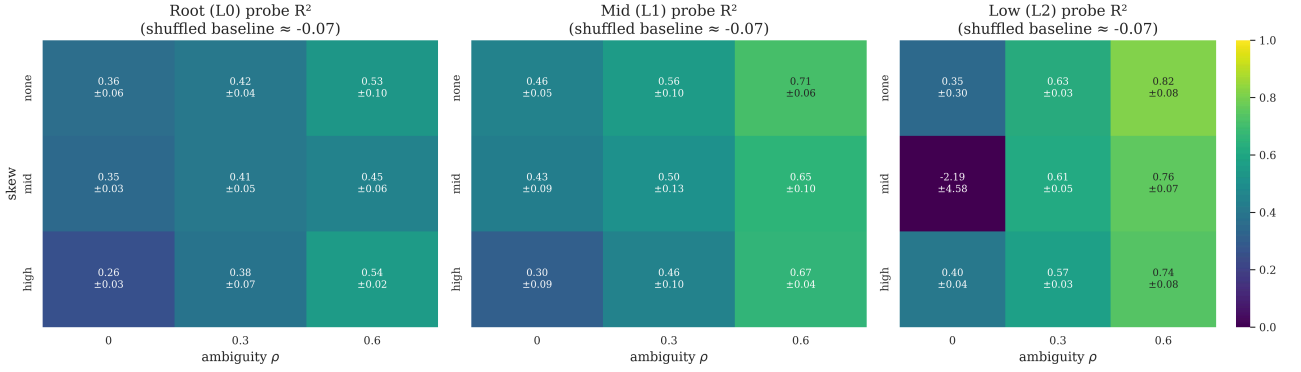


Figure 18: The linear probe survives ambiguity and skew, and **sharpens** with ambiguity. Per-level probe R^2 (root L_0 , mid L_1 , low L_2) over the 3×3 skew $\times$ ambiguity grid; each cell is mean $\pm$ SD over 3 rule-table draws, against a shuffled baseline of ≈ -0.07 . R^2 rises left-to-right (with ρ) at every level. The single dark L_2 cell at (mid, $\rho=\{= \}0$) is a near-deterministic-target instability, not a decodability failure.

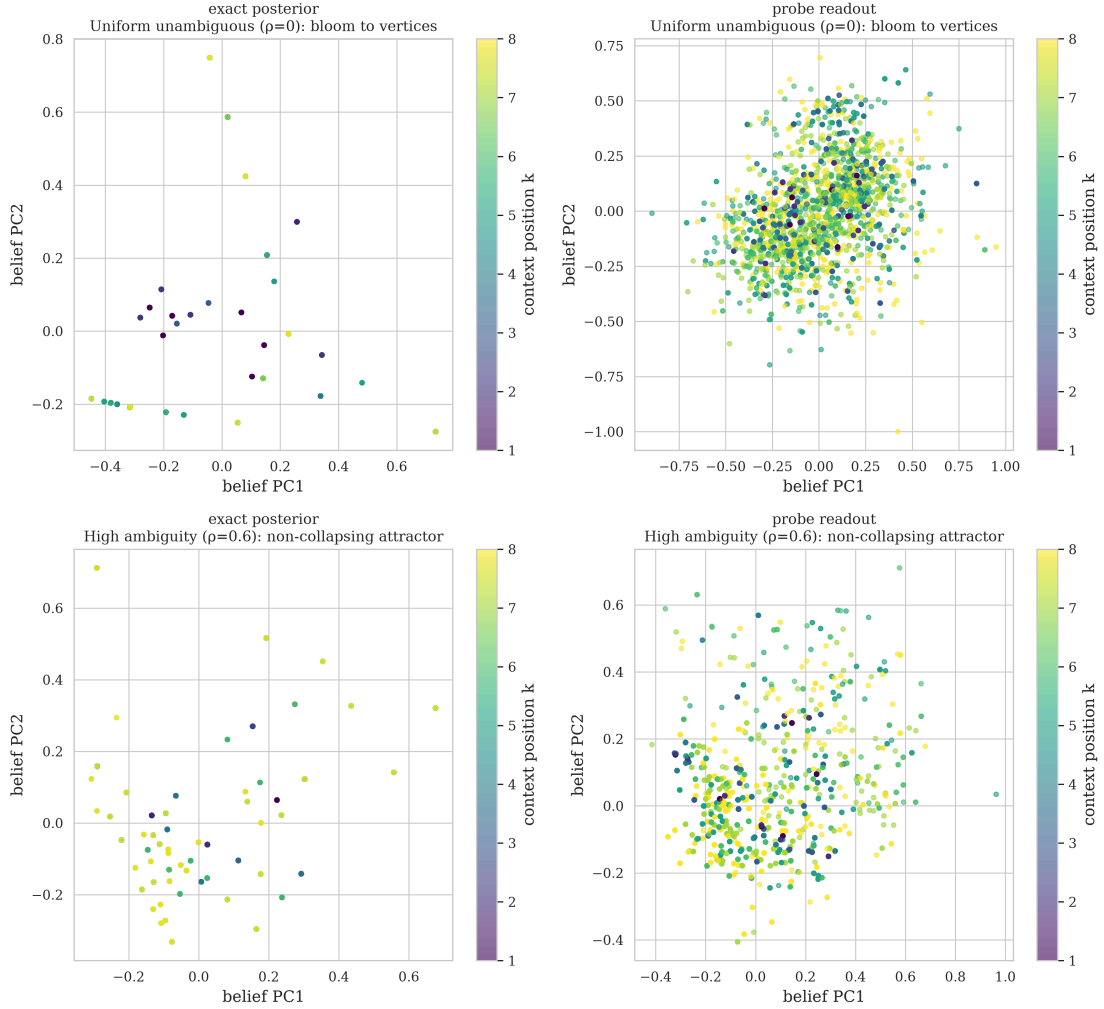


Figure 19: Exact posterior (left column) and linear-probe readout (right column) in belief-PCA, colored by context position, for the uniform corner $\rho = 0$ (top) and the high-ambiguity corner $\rho = 0.6$ (bottom). At $\rho = 0$ the late-context beliefs reach the simplex vertices (bloom-to-certainty); at $\rho = 0.6$ they form a structured interior cloud — the non-collapsing attractor — which the probe readout traces.

11. From fixed trees to recursive parse beliefs

The next step is not to make this RHM larger for its own sake, but to relax the assumption that makes it exactly solvable: the fixed tree. The present grammar is a finite, depth-limited PCFG whose hidden trees can be enumerated exactly. For $v = 8$, $s = 2$, $L = 3$, and $m = 2$, the tree has $1 + 2 + 4 = 7$ internal rule choices. Conditional on a root, this gives $2^7 = 128$ hidden trees; across the eight roots it gives $8 \cdot 128 = 1024$ total trees, and under the unambiguous-rule constraint these map one-to-one to leaf strings. The linguistic analogy is limited: a depth-3 binary RHM resembles a stylized binary phrase-structure grammar in branching structure, while attested grammars vary in depth, arity, vocabulary distribution, and rule multiplicity.

The fixed topology is the benchmark’s strength. It lets us run exact sum-product BP, enumerate all trees, count level-wise statistics cleanly, and avoid PCFG properness and normalization problems. It also removes parsing. Once the tree can vary in shape or depth, the learner must infer structure as well as symbols: inside-outside, Earley/Stolcke, or other chart parsers replace the fixed-tree BP pass. With unbounded recursion, the predictive state may require an unbounded stack and a countably infinite family of chart configurations. This is the real boundary between the current experiment and syntax-like recursion.

Cross that boundary gradually. Cap a recursive PCFG at a fixed depth so inference stays bounded. Keep the vocabulary small ($q = v = 8$), keep binary rules, reuse the same categories across depths, and impose a hard maximum derivation depth $D_{\max}$. Train the same scale of decoder transformer, alongside an LSTM and a stack-augmented baseline, on shallow depths and test on deeper held-out depths.

First run a Dyck-style bracket experiment to test stack tracking without lexical ambiguity. Compare the structures directly: right-branching or tail-recursive grammars should stress memory differently from center-embedded grammars, even when length, category inventory, and depth cap are matched.

This extension lets us compare HMM beliefs with grammar beliefs. In an HMM, the exact belief state is a point in a simplex over hidden states. In the fixed-tree RHM, the analogous object is the prefix-conditioned collection of BP node marginals and messages. The full joint belief over hidden trees is exponential, but the marginal targets have far fewer dimensions. In the $L = 3$ grammar, all node marginals occupy at most $7 \cdot (8 - 1) = 49$ coordinates; the ancestor path relevant to a particular next leaf occupies about $3 \cdot (8 - 1) = 21$. In a recursive PCFG, the natural targets become chart beliefs instead: span/category marginals and rule/split marginals computed by inside-outside inference.

The falsifiable question is not which metaphor fits, but what belief object the model linearizes. Does the residual stream encode the exponential joint parse posterior, or a factored direct sum of local chart beliefs? If it is factored, do span, category, and split beliefs occupy separable or approximately orthogonal subspaces? The fixed-tree results suggest a concrete prior: local, prediction-adjacent beliefs may be easier to read than the global root. Recursive grammars test whether that inverted strength gradient survives when the model must maintain alternative parses instead of filling a known tree template.

This program is not a search for fractal geometry in the present RHM. Fixed-depth RHM beliefs form a finite set, and bounded recursion remains finite once $D_{\max}$ is fixed. What the cap can reveal is the deformation of the representation as the allowed stack grows: whether linear decodability degrades smoothly, whether chart-like subspaces appear, and whether center embedding produces a different geometry from tail recursion. Claims about self-similar belief sets require a different regime — non-unifilar processes, infinite memory, or genuinely unbounded recursion.

12. Discussion and limitations

The picture is coherent: a tiny transformer trained to near-Bayes-optimal loss on a hierarchical grammar carries a **partial** linear image of the posterior over the grammar’s hidden latents in its residual stream — fuzzy for the coarse root ($R^2 = 0.38$), sharper for the local latents (R^2 up to 0.66). That image is assembled additively across layers, blooms from the prior toward the vertices as evidence accumulates, spans the full latent hierarchy, and is causally relied upon at generation time. We probe against an **exactly known** belief state (computed by belief propagation, not approximated), which is what lets us quantify the recovery as partial rather than merely assert it — but the linear recovery itself is imperfect, and we do not claim the probe reconstructs the posterior exactly. This reproduces the core Simplex belief-geometry phenomenology in a setting where the ground-truth belief state is known exactly.

Methodologically, this paper sits between the two reference points in the bibliography: Cagnetta et al.’s RHM study uses the same hierarchical data model but evaluates the last-token prediction setting, while Shai et al.’s **Transformers learn factored representations** motivates pooling predictive vectors across contexts. Our readout follows the latter all-context-position spirit, treating every prefix position as a belief state to be decoded, while keeping Cagnetta et al.’s exact RHM grammar as the data source.

The most interesting wrinkle is the **inverted strength gradient**: the global root, the spec’s nominal target, is the hardest latent to decode, while local near-leaf latents are read off most cleanly. This is intuitive in hindsight — next-token prediction rewards representing whatever is most locally predictive — but it sharpens the belief-geometry claim: the residual stream tracks the **full** latent posterior, weighted toward what the task needs, not a single privileged variable.

Limitations. (i) The grammar is deliberately tiny and fully enumerable ($s = 2$, $L = 3$, 1024 trees); this is what makes the beliefs exact and the verification airtight, but it leaves open how the geometry scales to larger, non-enumerable RHMs. (ii) At $L = 3$ the root posterior collapses to certainty by $k = 3$ on many strings, compressing the dynamic range over which blooming is visible for the root; the spec anticipated this and suggested $L = 4$ as a follow-up. (iii) The strongly-negative mid-layer probe cells indicate the affine probe is locally mis-specified at some position/layer combinations; a per-position or whitened probe would tighten these estimates. (iv) Steering is applied at a single layer (layer 1) along a mean-difference axis; a learned causal direction and a layer sweep would strengthen the causal claim. (v) The detailed

pass-1 figures (Figure 2–Figure 6) are from a single training run and grammar draw; the grammar-sweep and architecture-sweep sections quantify how the **headline** metrics move across 10 grammars / 3 seeds and across 11 capacity configs / 3 grammars / 3 seeds respectively, but the architecture sweep is one-axis-at-a-time (no $n_{\text{layer}} \times n_{\text{embd}} \times n_{\text{head}}$ interaction cells). The $L = 4$ section extends the family one tree level deeper; still-larger grammars ($L \geq 5$, larger v , $m > 2$), where brute-force belief verification becomes intractable, remain a future pass.

13. Conclusion

On an exactly-solvable hierarchical grammar, a small transformer’s residual stream is a linear image of the exact Bayesian belief simplex: it accumulates across depth, blooms with context, encodes the whole latent hierarchy (local latents most strongly), and is causally used. Pre-registered predictions held in three of four cases, and the one miss — a weaker root probe than predicted — turned out to be a feature of the hierarchy rather than a failure of the belief-geometry hypothesis. The exact, enumerable setting turns a qualitative interpretability story into quantitative, falsifiable measurement.

14. Appendix: Root reconstruction diagnostics

The root-posterior probe is a regression target, but it is also useful to ask the harsher classification question: does the largest coordinate of the linear belief readout recover the sampled root class? Across 400 probe examples and 8 context positions (3200 example-position points), the answer is yes for 1565 points (48.91%). This is well above random guessing, but random guessing is only $1/8 = 12.5\%$ because there are eight root classes; the relevant baseline is not 50%.

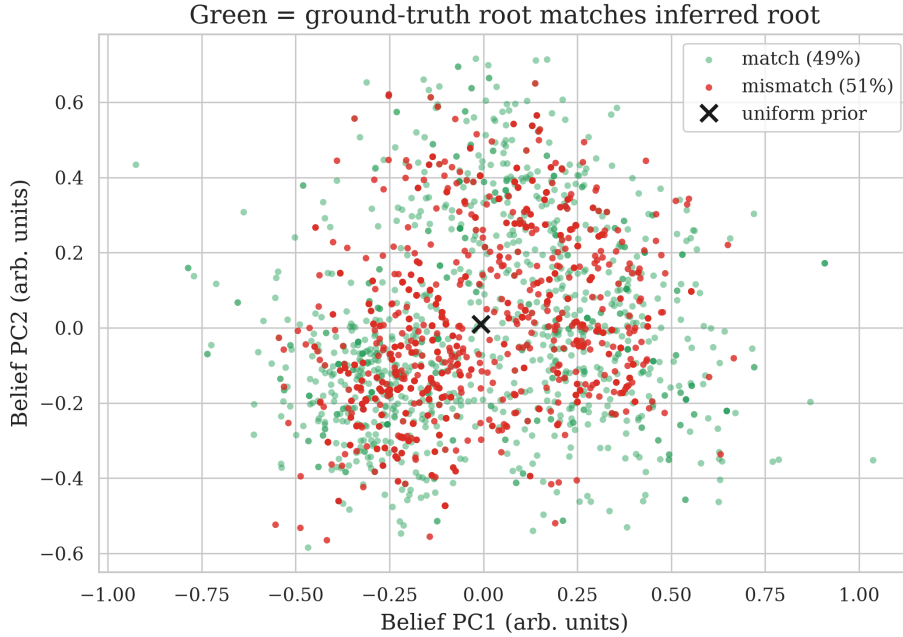


Figure 20: Root reconstruction from the linear belief readout. Green points are example-position pairs where the readout’s argmax matches the ground-truth root; red points are mismatches. The overall match rate is 48.91%, compared with a random-guessing baseline of 12.5% (not 50%) for eight root classes.

Restricting to positions where nearly all evidence is visible makes the diagnostic sharper. With seven of eight symbols observed ($k = 6$), the exact Bayesian posterior’s MAP root already matches the sampled root in 95.00% of examples, while the linear readout matches in 62.25%. With all eight symbols observed ($k = 7$), the exact posterior is deterministic, but the readout still reaches only 70.50%. Thus early ambiguity explains part, but not all, of the fuzzy root reconstruction.

Subset	Prefix length	Readout match	Exact MAP match
All example-position points	1-8	$\frac{1565}{3200} = 48.91\%$	—
All but last symbol	7	$\frac{249}{400} = 62.25\%$	$\frac{380}{400} = 95.00\%$

Subset	Prefix length	Readout match	Exact MAP match
Full sequence	8	$\frac{282}{400} = 70.50\%$	$\frac{400}{400} = 100.00\%$

15. Appendix: Per-run grammar-sweep sanity table

Every run in the grammar sweep, no filtering. `grammar` indexes the rule-table draw (`Grammar.random(seed=grammar)`); `seed` is the master replicate seed; the architecture is pinned. `test_ce` is the held-out next-token cross-entropy; `loss_gap_closed` is the fraction of the uniform→Bayes gap closed (sanity covariate, not a gate); `root_r2` and `deepest_r2` are the level- L_0 and mean level- L_2 probe R^2 . Loaded directly from `figures/sweep_sanity.csv`.

grammar	seed	n_layer	n_embd	n_head	test_ce	loss_gap_closed	root_r2	deepest_r2
0	0	2	128	4	0.931	0.848	0.242	0.498
0	1	2	128	4	0.891	0.878	0.243	0.497
0	2	2	128	4	0.918	0.858	0.281	0.532
1	0	2	128	4	0.847	0.906	0.375	0.556
1	1	2	128	4	0.812	0.932	0.479	0.616
1	2	2	128	4	0.845	0.908	0.353	0.496
2	0	2	128	4	0.905	0.886	0.402	-0.061
2	1	2	128	4	0.858	0.922	0.454	0.479
2	2	2	128	4	0.866	0.916	0.354	0.46
3	0	2	128	4	0.845	0.91	0.336	0.253
3	1	2	128	4	0.912	0.86	0.333	0.611
3	2	2	128	4	0.92	0.854	0.311	0.607
4	0	2	128	4	0.925	0.876	0.442	0.505
4	1	2	128	4	0.884	0.907	0.416	0.497
4	2	2	128	4	0.909	0.888	0.29	0.442
5	0	2	128	4	0.853	0.911	0.25	0.461
5	1	2	128	4	0.841	0.92	0.277	0.482
5	2	2	128	4	0.848	0.914	0.334	0.493
6	0	2	128	4	0.868	0.896	0.351	0.523
6	1	2	128	4	0.882	0.885	0.4	0.553
6	2	2	128	4	0.836	0.919	0.312	0.523
7	0	2	128	4	0.964	0.82	0.298	0.441
7	1	2	128	4	0.909	0.861	0.256	0.359
7	2	2	128	4	0.892	0.874	0.326	0.436
8	0	2	128	4	0.897	0.863	0.338	0.524
8	1	2	128	4	0.838	0.906	0.359	0.532
8	2	2	128	4	0.868	0.884	0.327	0.532
9	0	2	128	4	0.899	0.894	0.411	0.557
9	1	2	128	4	0.988	0.827	0.394	0.565
9	2	2	128	4	0.87	0.916	0.438	0.604

Table 4: All 30 grammar-sweep runs (10 grammars $\times$ 3 seeds), pinned architecture, full 4000-step budget. No convergence filtering.

16. Appendix: Per-run architecture-sweep sanity table

Every run in the architecture sweep, no filtering — all 99 (11 OAT capacity configs $\times$ 3 grammars $\times$ 3 seeds). `n_layer`, `n_embd`, `n_head`, `steps` give the capacity config; `grammar` and `seed` the replicate. `test_ce` is the held-out next-token cross-entropy; `loss_gap_closed` the fraction of the uniform→Bayes gap closed (covariate, not a gate); `root_r2` and `deepest_r2` the level- L_0 and mean level- L_2 probe R^2 . The deliberately-underpowered small models (e.g. $n_{\text{embd}} = 16$) are included. Loaded directly from `figures/arch_sanity.csv`.

grammar	seed	n_layer	n_embd	n_head	steps	test_ce	loss_gap_closed	root_r2	deepest_r2
0	0	1	128	4	4000	0.974	0.816	0.136	0.374
0	1	1	128	4	4000	0.933	0.847	0.131	0.403
0	2	1	128	4	4000	1.028	0.776	0.114	0.354
1	0	1	128	4	4000	0.913	0.858	0.271	0.475
1	1	1	128	4	4000	0.873	0.887	0.268	0.483
1	2	1	128	4	4000	0.854	0.901	0.245	0.421
2	0	1	128	4	4000	0.92	0.875	0.292	-0.138
2	1	1	128	4	4000	0.872	0.912	0.248	0.344
2	2	1	128	4	4000	0.911	0.882	0.298	0.382
0	0	2	16	4	4000	1.065	0.749	0.022	0.172
0	1	2	16	4	4000	1.048	0.762	0.045	0.203
0	2	2	16	4	4000	1.038	0.769	0.032	0.183
1	0	2	16	4	4000	0.966	0.819	0.065	0.233
1	1	2	16	4	4000	0.951	0.83	0.079	0.216
1	2	2	16	4	4000	0.921	0.852	0.073	0.222
2	0	2	16	4	4000	0.943	0.857	0.097	-0.115
2	1	2	16	4	4000	0.973	0.835	0.08	0.096
2	2	2	16	4	4000	0.934	0.864	0.107	0.12
0	0	2	64	4	4000	1.107	0.718	0.143	0.329
0	1	2	64	4	4000	1.236	0.623	0.127	0.358
0	2	2	64	4	4000	1.225	0.631	0.151	0.351
1	0	2	64	4	4000	1.05	0.757	0.258	0.397
1	1	2	64	4	4000	1.182	0.66	0.225	0.381
1	2	2	64	4	4000	1.084	0.732	0.212	0.343
2	0	2	64	4	4000	1.182	0.677	0.249	-0.121
2	1	2	64	4	4000	1.122	0.722	0.283	0.304
2	2	2	64	4	4000	1.103	0.737	0.258	0.296
0	0	2	128	1	4000	0.919	0.857	0.205	0.458
0	1	2	128	1	4000	0.936	0.844	0.165	0.468
0	2	2	128	1	4000	1.07	0.745	0.155	0.432
1	0	2	128	1	4000	0.931	0.844	0.266	0.527
1	1	2	128	1	4000	0.943	0.836	0.274	0.512
1	2	2	128	1	4000	0.931	0.845	0.287	0.495
2	0	2	128	1	4000	0.916	0.878	0.369	0.091
2	1	2	128	1	4000	0.937	0.862	0.322	0.449
2	2	2	128	1	4000	1.02	0.799	0.247	0.371
0	0	2	128	2	4000	0.891	0.878	0.288	0.508
0	1	2	128	2	4000	0.871	0.893	0.269	0.594
0	2	2	128	2	4000	0.899	0.871	0.17	0.465
1	0	2	128	2	4000	0.856	0.9	0.449	0.585
1	1	2	128	2	4000	0.81	0.934	0.432	0.619
1	2	2	128	2	4000	0.867	0.892	0.344	0.495
2	0	2	128	2	4000	1.023	0.797	0.37	-0.19
2	1	2	128	2	4000	0.877	0.908	0.353	0.453
2	2	2	128	2	4000	0.859	0.921	0.365	0.477
0	0	2	128	4	4000	1.019	0.783	0.182	0.409
0	1	2	128	4	4000	0.964	0.824	0.214	0.498
0	2	2	128	4	4000	0.877	0.888	0.301	0.515
1	0	2	128	4	4000	0.89	0.875	0.467	0.576
1	1	2	128	4	4000	0.832	0.918	0.396	0.592
1	2	2	128	4	4000	0.819	0.927	0.405	0.54
2	0	2	128	4	4000	0.837	0.938	0.418	-0.028
2	1	2	128	4	4000	0.935	0.864	0.37	0.404
2	2	2	128	4	4000	0.883	0.903	0.426	0.464
0	0	2	128	4	16000	1.307	0.571	0.304	0.477
0	1	2	128	4	16000	1.17	0.671	0.298	0.566
0	2	2	128	4	16000	1.278	0.592	0.304	0.517
1	0	2	128	4	16000	1.086	0.73	0.406	0.558
1	1	2	128	4	16000	1.074	0.739	0.442	0.603
1	2	2	128	4	16000	0.972	0.814	0.474	0.583
2	0	2	128	4	16000	0.984	0.827	0.497	-0.171
2	1	2	128	4	16000	1.096	0.742	0.478	0.552
2	2	2	128	4	16000	0.895	0.894	0.448	0.506
0	0	2	128	8	4000	1.046	0.763	0.253	0.461

Table 5: All 99 architecture-sweep runs (11 one-axis-at-a-time capacity configs $\times$ 3 grammars $\times$ 3 seeds).

No convergence filtering; small models are expected to underfit.

17. Appendix: Per-run $L = 4$ depth-sweep sanity table

Every run in the $L = 4$ depth sweep, no filtering — all 60 (10 grammars $\times$ 3 seeds $\times$ $n_{\text{layer}} \in \{2, 3\}$). `grammar` and `seed` index the rule-table draw and master replicate seed; `n_layer` is the only varied architecture axis ($n_{\text{embd}} = 128$, $n_{\text{head}} = 4$, 4000 steps throughout). `test_ce` is the held-out next-token cross-entropy; `loss_gap_closed` the fraction of the uniform $\rightarrow$ Bayes gap closed (covariate, not a gate); `root_r2` and `deepest_r2` the level- L_0 and mean leaf-parent- L_3 probe R^2 . Loaded directly from `figures/depth4_sanity.csv`.

grammar	seed	n_layer	test_ce	loss_gap_closed	root_r2	deepest_r2
0	0	2	0.716	0.992	0.207	0.32
0	0	3	0.716	0.991	0.23	0.35
0	1	2	0.715	0.992	0.178	0.345
0	1	3	0.719	0.99	0.263	0.386
0	2	2	0.728	0.983	0.148	0.307
0	2	3	0.711	0.995	0.276	0.33
1	0	2	0.728	0.982	0.227	0.284
1	0	3	0.713	0.993	0.282	0.298
1	1	2	0.715	0.992	0.227	0.28
1	1	3	0.718	0.989	0.288	0.341
1	2	2	0.722	0.987	0.179	0.273
1	2	3	0.715	0.992	0.298	0.336
2	0	2	0.724	0.987	0.203	0.317
2	0	3	0.726	0.986	0.271	0.372
2	1	2	0.727	0.985	0.171	0.304
2	1	3	0.718	0.992	0.301	0.384
2	2	2	0.725	0.986	0.149	0.315
2	2	3	0.72	0.99	0.318	0.386
3	0	2	0.74	0.986	0.231	0.25
3	0	3	0.732	0.992	0.304	0.235
3	1	2	0.783	0.955	0.22	0.257
3	1	3	0.73	0.994	0.351	0.244
3	2	2	0.75	0.979	0.211	0.273
3	2	3	0.735	0.99	0.345	0.301
4	0	2	0.73	0.987	0.183	0.381
4	0	3	0.722	0.992	0.33	0.423
4	1	2	0.723	0.992	0.171	0.347
4	1	3	0.731	0.986	0.2	0.368
4	2	2	0.722	0.992	0.244	0.388
4	2	3	0.721	0.993	0.308	0.443
5	0	2	0.745	0.978	0.073	0.273
5	0	3	0.732	0.988	0.134	0.288
5	1	2	0.741	0.981	0.103	0.317
5	1	3	0.728	0.991	0.205	0.33
5	2	2	0.748	0.976	0.134	0.283
5	2	3	0.743	0.98	0.153	0.306
6	0	2	0.718	0.992	0.213	0.31
6	0	3	0.72	0.991	0.29	0.386
6	1	2	0.731	0.983	0.134	0.29
6	1	3	0.731	0.982	0.206	0.351
6	2	2	0.723	0.988	0.138	0.299
6	2	3	0.717	0.993	0.266	0.34
7	0	2	0.732	0.987	0.254	0.259
7	0	3	0.744	0.978	0.228	0.274
7	1	2	0.729	0.989	0.227	0.103
7	1	3	0.762	0.965	0.249	0.121
7	2	2	0.732	0.987	0.186	-0.229
7	2	3	0.737	0.983	0.245	-0.216
8	0	2	0.718	0.988	0.202	0.292
8	0	3	0.714	0.99	0.257	0.33
8	1	2	0.727	0.981	0.236	0.276
8	1	3	0.713	0.991	0.294	0.315
8	2	2	0.732	0.978	0.152	0.245
8	2	3	0.718	0.988	0.327	0.347
9	0	2	0.727	0.986	0.27	0.298
9	0	3	0.727	0.986	0.383	0.348
9	1	2	0.721	0.991	0.264	0.328
9	1	3	0.716	0.994	0.379	0.387
9	2	2	0.717	0.993	0.298	0.326
9	2	3	0.724	0.988	0.333	0.351

Table 6: All 60 $L = 4$ depth-sweep runs ($10 \text{ grammars} \times 3 \text{ seeds} \times n_{\text{layer}} \in \{2, 3\}$). No convergence filtering. **deepest_r2** is the mean over the eight leaf-parent (L_3) nodes.

18. Appendix: Per-run non-collapse sweep sanity table

Every run in the non-collapse sweep, no filtering — all 27 ($3 \text{ skew} \times 3 \text{ ambiguity} \times 3 \text{ rule-table draws}$), at the pinned arch ($n_{\text{layer}} = 2, n_{\text{embd}} = 128, n_{\text{head}} = 4, 4000 \text{ steps}$). `bayes_floor` is the exact weighted Bayes-optimal next-token entropy; `loss_gap_closed` the fraction of the uniform→Bayes gap closed (covariate); `root_r2/mid_r2/low_r2` the level- $L_0/L_1/L_2$ probe R^2 (shuffled baseline shuffled_r2); `k8_entropy` the headline mean $k = 8$ root posterior entropy; `eff_dim/hull_area` the reachable-set descriptors. Loaded directly from `figures/noncollapse_sanity.csv`.

skew	ambiguity	draw	test_ce	bayes_floor	loss_gap_closed	root_r2	mid_r2	low_r2	shuffled_r2	k8_entropy	eff_dim	hull_area
high	0.0000	0	1.1915	0.2818	0.4939	0.2369	0.3522	0.4214	-0.0868	0.0000	6.9652	0.6465
high	0.0000	1	0.8751	0.3953	0.7151	0.2492	0.3554	0.4202	-0.0628	0.0000	6.9652	0.6465
high	0.0000	2	1.0670	0.2532	0.5544	0.2983	0.1920	0.3542	-0.0782	0.0000	6.9652	0.6465
high	0.3000	0	0.9041	0.2118	0.6293	0.3115	0.3577	0.5997	-0.0447	0.2655	5.6327	0.5984
high	0.3000	1	0.3135	0.1622	0.9211	0.4511	0.5554	0.5639	-0.0808	0.3314	4.9094	0.6720
high	0.3000	2	0.6246	0.2468	0.7938	0.3783	0.4539	0.5344	-0.1115	0.4441	4.7952	0.6780
high	0.6000	0	0.3267	0.1826	0.9241	0.5202	0.6696	0.6840	-0.0675	0.5852	3.7205	0.6501
high	0.6000	1	0.2625	0.2101	0.9720	0.5496	0.6274	0.8281	-0.0704	0.6123	3.6509	0.7863
high	0.6000	2	0.2848	0.2101	0.9601	0.5612	0.7059	0.7040	-0.1028	0.4243	2.8199	0.6248
mid	0.0000	0	0.8921	0.5366	0.7696	0.3287	0.4008	0.3964	-0.0608	0.0000	6.9652	0.6465
mid	0.0000	1	1.0199	0.5468	0.6913	0.3799	0.5235	0.5137	-0.0439	0.0000	6.9652	0.6465
mid	0.0000	2	1.1025	0.5426	0.6357	0.3549	0.3549	-7.4713	-0.0763	0.0000	6.9652	0.6465
mid	0.3000	0	0.6272	0.4867	0.9118	0.4159	0.4348	0.5595	-0.0552	0.6537	4.8768	0.5921
mid	0.3000	1	0.6823	0.4995	0.8843	0.4557	0.6463	0.6656	-0.0553	0.3870	5.8288	0.6014
mid	0.3000	2	0.6325	0.5326	0.9354	0.3626	0.4213	0.6025	-0.0270	0.7169	5.2545	0.7321
mid	0.6000	0	0.5376	0.4261	0.9326	0.5044	0.6875	0.8028	-0.0715	1.1605	4.3417	0.5354
mid	0.6000	1	0.6313	0.5088	0.9220	0.3874	0.5424	0.6770	-0.0920	0.9423	3.8371	0.6337
mid	0.6000	2	0.5529	0.4420	0.9323	0.4711	0.7290	0.7901	-0.1003	1.0470	4.0308	0.4432
none	0.0000	0	0.8900	0.7253	0.8784	0.3002	0.4092	0.5039	-0.0568	0.0000	6.9652	0.6465
none	0.0000	1	0.8494	0.7199	0.9047	0.3578	0.5082	0.5504	-0.0638	0.0000	6.9652	0.6465
none	0.0000	2	0.8447	0.7544	0.9319	0.4262	0.4582	0.0083	-0.0640	0.0000	6.9652	0.6465
none	0.3000	0	0.6930	0.6221	0.9513	0.4455	0.4894	0.6512	-0.0439	0.7598	5.3126	0.6421
none	0.3000	1	0.7216	0.6509	0.9505	0.4274	0.6678	0.6417	-0.0553	0.6576	5.5352	0.5416
none	0.3000	2	0.7397	0.6660	0.9479	0.3771	0.5106	0.5869	-0.0733	0.7003	5.4769	0.5762
none	0.6000	0	0.5011	0.4860	0.9905	0.5640	0.6747	0.7615	-0.1606	1.3969	3.7241	0.6400
none	0.6000	1	0.5658	0.5448	0.9863	0.6170	0.7852	0.9018	-0.0745	1.3457	3.3016	0.5520
none	0.6000	2	0.5328	0.4976	0.9778	0.4220	0.6758	0.7826	-0.0481	1.4068	2.7883	0.4318

Table 7: All 27 non-collapse runs ($3 \text{ skew} \times 3 \text{ ambiguity} \times 3 \text{ draws}$). No filtering. `k8_entropy` ≈ 0 exactly when $\rho = 0$ regardless of skew.

19. References

Cagnetta, F., Favero, A., Sclocchi, A., and Wyart, M. (2025). **Scaling Laws and Representation Learning in Simple Hierarchical Languages: Transformers vs. Convolutional Architectures**. arXiv:2505.07070.

Shai, A. et al. (2026). **Transformers learn factored representations**. arXiv:2602.02385.

Project sources: `spec.md`, `spec-grammar-sweep.md`, `PREREGISTRATION.md`, `EXECUTION_OUTPUT.md`, `results/analysis.json`, `results/root_reconstruction.json`, `artifacts/train_summary.json`. Grammar sweep: `sweep.py`, `sweep.yaml`, `results/sweep/g*/{config.json,analysis.json}`, `figures/sweep_sanity.csv`. Architecture sweep: `spec-arch-sweep.md`, `run_arch.py`, `results/arch/g*/{config.json,analysis.json}`, `figures/arch_sanity.csv`.